Social Media Analytics of Taylor Swift, Ariana Grande and Billie Eilish

3032ICT - Big Data and Social Media Analytics

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1. Artist Descriptions

Taylor Swift, Ariana Grande and Billie Eilish are pop artists with different career lengths, catalogues and musical approaches. The comparison uses the catalogue inventory recorded on 22 September 2026.

Artist	Years active	Studio albums	Released songs
Taylor Swift	2003-2026: about 23 years	12 + 4 re-recorded albums	283
Ariana Grande	2008-2026: about 18 years	8	212
Billie Eilish	2015-2026: about 11 years	3	66

Table 1. Career length and catalogue comparison

Counting rules. Career duration is 2026 minus the documented start year, not time since the first album ("Ariana Grande," n.d.; "List of Songs Recorded by Billie Eilish," n.d.; "Taylor Swift," n.d.). Original studio albums are counted; EPs, live albums and soundtrack compilations are excluded. Taylor's re-recordings are separate. Ariana's eighth album, *Petal*, was released in July 2026 (Universal Music Canada, 2026).

Song totals count distinct released titles, including album/bonus tracks, covers, featured or cast performances and named intros/interludes. Re-recordings and alternate versions count once; unreleased songs, writing-only credits and backing-vocal-only appearances are excluded. These are audited inventory counts, not official lifetime totals. The three song lists, adjustments and limitations are documented on page 26 ("List of Songs by Taylor Swift," n.d.; "List of Songs Recorded by Ariana Grande," n.d.; "List of Songs Recorded by Billie Eilish," n.d.).

Career and catalogue comparison

Taylor has the longest career and largest catalogue, giving audiences more musical eras to discuss. Ariana is intermediate and also works in theatre and film. Billie has the shortest career and fewest albums. Catalogue size gives context but does not establish popularity or explain differences in these equally sized comment samples.

Collaborators

Taylor regularly works with Jack Antonoff, Aaron Dessner, Max Martin and Shellback; featured performers include Ed Sheeran and Post Malone. Ariana works with ILYA, Max Martin, Tommy Brown and Victoria Monét, with vocal collaborations including Nicki Minaj and The Weeknd. Billie writes and records mainly with Finneas; collaborators include Khalid, Rosalía and Charli XCX. Taylor and Ariana use broader recurring teams, while Billie has a concentrated core partnership ("List of Songs by Taylor Swift," n.d.; "List of Songs Recorded by Ariana Grande," n.d.; "List of Songs Recorded by Billie Eilish," n.d.; Universal Music Canada, 2026).

Additional characteristic 1: Musical style and development

Taylor moved from country to pop and explored indie-folk sounds on *Folklore* and *Evermore*. Ariana combines pop and R&B with dance influences. Billie uses electropop, restrained vocals and darker themes. Taylor shows the clearest movement between genres, while Ariana's vocal pop/R&B and Billie's quieter delivery offer contrasting listening experiences ("Ariana Grande," n.d.; "List of Songs Recorded by Billie Eilish," n.d.; Recording Academy, n.d.-b).

Additional characteristic 2: Awards and achievements

Taylor has 14 Grammy wins, including a record four Album of the Year awards. Ariana has three, including the 2026 Pop Duo/Group award with Cynthia Erivo for "Defying Gravity." Billie has ten; she and Finneas became the first three-time Song of the Year winners with "Wildflower" in 2026. Taylor's distinction is repeated album recognition, Ariana's includes collaborative vocal performance, and Billie's songwriting record is notable within a shorter career. Awards measure industry recognition rather than every fan's opinion (Guinness World Records, n.d.; Recording Academy, n.d.-a, n.d.-b).

2. YouTube Data Collection

Collection method

The data were collected from YouTube with R, the tuber package and the YouTube Data API. A fixed publication period from 1 January 2024 to 23 August 2026 was used. Searches were limited to official artist and VEVO channels where possible.

The search strategy used several phrases, including official music video, lyric video, official audio and live performance. Duplicates were removed by video ID. Fan uploads, trailers, behind-the-scenes videos and other unrelated material were excluded. Nine suitable videos were then selected for each artist.

The retained sample contains 400 comment records per video: 3,600 per artist and 10,800 overall. This cap was applied after retrieval, including any replies returned. Equal counts simplify comparison but do not make the sample representative or preserve complete discussion threads.

Artist	Videos	Comments per video	Comments
Taylor Swift	9	400	3600
Ariana Grande	9	400	3600
Billie Eilish	9	400	3600
Total	27	-	10800

Table 2. Final sample size

Selected videos

Artist	Video title	Search category
Taylor Swift	The Fate of Ophelia (Official Music Video)	Official music video
Taylor Swift	Fortnight (feat. Post Malone) (Official Music Video)	Official music video
Taylor Swift	I Can Do It With a Broken Heart (Official Video)	Official music video
Taylor Swift	Opalite (Official Music Video)	Official music video
Taylor Swift	Elizabeth Taylor (Official Music Video)	Official music video
Taylor Swift	Opalite (Lyric Video)	Lyric video
Taylor Swift	So High School (Official Lyric Video)	Lyric video
Taylor Swift	The Fate of Ophelia (Lyric Video)	Lyric video
Taylor Swift	I Hate It Here (Official Lyric Video)	Lyric video
Ariana Grande	yes, and? (official music video)	Official music video
Ariana Grande	the boy is mine (Official Music Video)	Official music video
Ariana Grande	supernatural (official music video)	Official music video
Ariana Grande	we can't be friends (wait for your love) (official music video)	Official music video
Ariana Grande	hate that i made you love me (official music video)	Official music video
Ariana Grande	hate that i made you love me (official lyric video)	Lyric video
Ariana Grande	supernatural (lyric visualizer)	Lyric video
Ariana Grande	petal (official lyric video)	Lyric video
Ariana Grande	twilight zone (lyric visualizer)	Lyric video
Billie Eilish	WILDFLOWER (Live Performance from Amazon Music's Songline)	Official video search

Artist	Video title	Search category
Billie Eilish	BIRDS OF A FEATHER (Live Performance from Amazon Music's Songline)	Official video search
Billie Eilish	BIRDS OF A FEATHER (Live from the LA28 Olympic Handover)	Official video search
Billie Eilish	WILDFLOWER (COLORS Performance)	Official video search
Billie Eilish	L'AMOUR DE MA VIE [OVER NOW EXTENDED EDIT]	Official video search
Billie Eilish	What Was I Made For? (From the Motion Picture Barbie) [The Barbie Montage]	Official video search
Billie Eilish	L'AMOUR DE MA VIE (Live Performance from Amazon Music's Songline)	Official video search
Billie Eilish	THE GREATEST (Live from the LA28 Olympic Handover)	Live performance
Billie Eilish	WILDFLOWER (Live from FireAid, 2025)	Live performance

Table 3. Twenty-seven selected YouTube videos

Cleaning and validation

The first validation found ten blank texts across eight videos. Those videos were retrieved again with a limit of 450; blank texts and duplicate comment IDs were removed, then the first 400 valid rows were retained. Collection dates were 22 August and 30 August 2026. The publication window applies to videos, not comment dates.

Validation measure	Result
Clean comments	10800
Unique comment IDs	10800
Blank comment text	0
Artists	3
Videos	27
Minimum comments per video	400
Maximum comments per video	400

Table 4. Final data validation

Sampling limitation and replies. The fixed 400-row cap discarded extra records. The available first raw download contains 948 replies within its 10,800 rows, so the cap does not mean 400 top-level comments plus their replies. Because eight videos were later replaced, that raw reply count must not be presented as the final clean dataset's reply total.

The final clean export does not retain parent-comment IDs, so its complete thread structure cannot be verified. The results below therefore describe the existing capped sample. An improved collection would set a top-level target, retain all available replies, and report the actual counts per artist. That recollection has not been completed for this report.

The selection also differs in video format: Taylor and Ariana include lyric videos, while Billie includes several live performances. Lyrics, performance comments, collection timing and the cap may all affect differences between artists.

2. YouTube Data Collection: Search Audit

The saved R script makes the search and selection rules reproducible. Every query used video-publication bounds of 1 January 2024 at 00:00 UTC and before 23 August 2026 at 00:00 UTC, with `relevance_language = "en"`. This requests English relevance; it does not restrict all returned comments to English. Each query below begins with the artist name.

Artist	Query suffixes and requested limits	Channel handling	Final selection
Taylor Swift	official music video: 10 lyric video: 10 official audio: 10	Music results filtered to Taylor Swift. Lyric/audio requests use the identified artist channel.	5 music-video results; 4 lyric results
Ariana Grande	official music video: 10 lyric video: 10 official audio: 10	Requests restricted to the ArianaGrandeVevo channel.	5 music-video results; 4 lyric results
Billie Eilish	official music video: 50 lyric video: 50 official audio: 50 live performance: 20	First three result sets filtered to artist/VEVO channel IDs. Live search restricted to the main artist channel.	7 official-video-search results; 2 live-search results

Table 4a. Recorded search design. Limits are requested maxima, not verified numbers of returned or unique videos.

Eligibility, ordering and stopping rule

Candidate videos were deduplicated by video ID, retaining the first search category encountered. For Taylor and Ariana, titles were checked for music-video, lyric or audio wording. Eligible music and lyric candidates were sorted by reported comment count; the five highest-comment music videos and four highest-comment lyric videos were selected. Audio was searched but did not enter the final sample.

For Billie, titles matching trailer, behind, story of my song, on tour or album cover were excluded. The nine highest-comment eligible candidates across the searched categories were retained. Eligibility required at least 400 reported comments. The stopping rule was a nine-video sample per artist; it was not exhaustion of all channel uploads.

Interpreting the search labels

Billie's seven "official-video-search" records include performances, an extended edit and a montage. That label records which query found a video, rather than proving that it is a conventional music video. The broader Billie searches increased candidate coverage but did not make the final video formats identical across artists.

Coverage and audit limits

The strategy covers several official release formats and screens for relevant artist uploads. It remains a bounded, purposive sample: query limits, channel filters and selection by comment count favour visible, active videos. Original search-hit, intermediate-candidate and exclusion counts were not retained, so a complete screening flow or exhaustive catalogue coverage cannot be claimed. The requested limits must not be reported as actual retrieved counts.

Evidence: 01_Code/Q2_Data_Collection..R and 03_Clean_Data/Q2_Selected_Videos.csv. The existing 400-row cap and reply limitations remain as described in Q2; no comments or results were changed for this audit.

Collection validation evidence

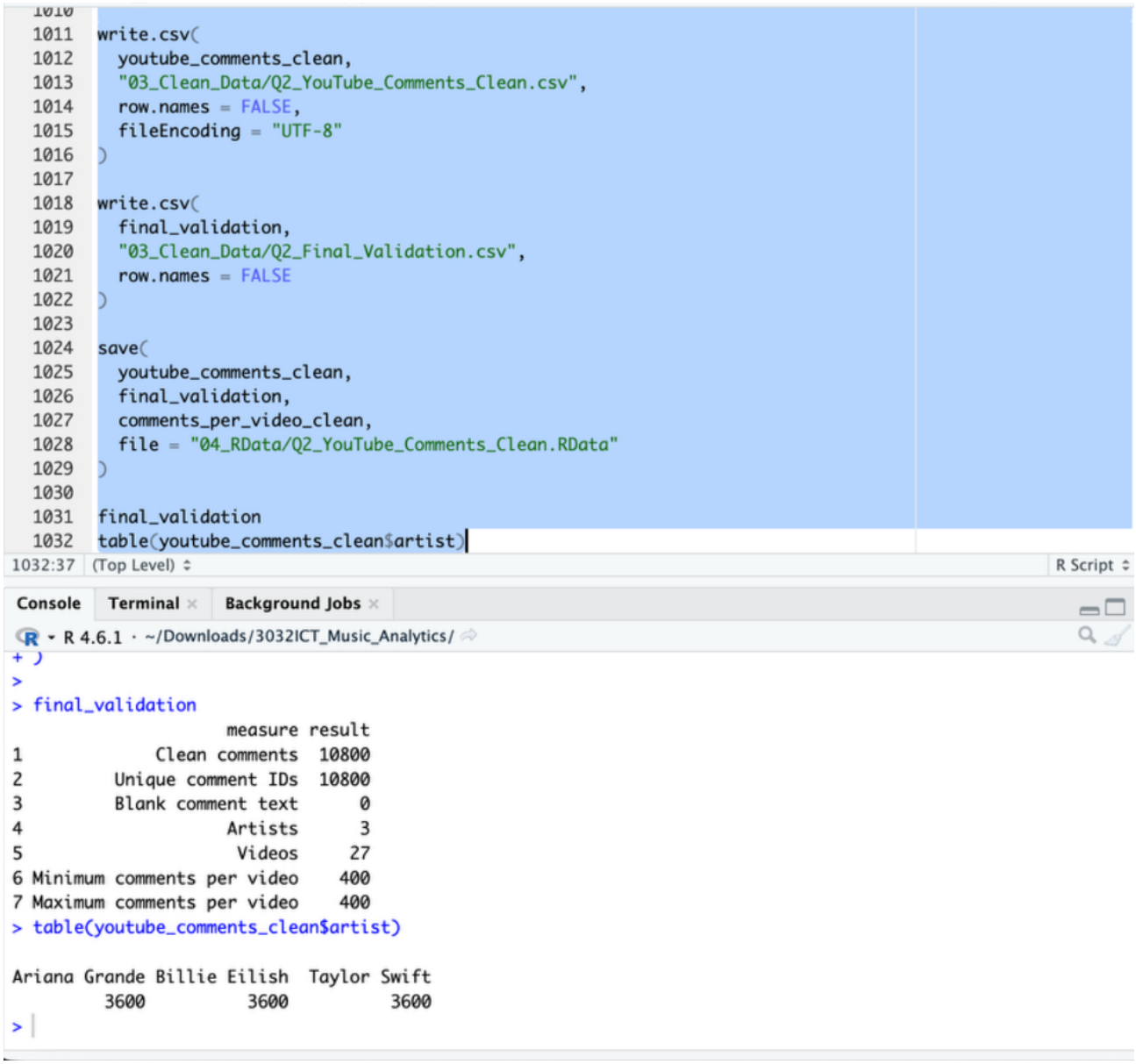


Figure 1. Original R code and validation output for the capped clean dataset; desktop controls cropped out.

3. Top Five Actors

An actor is a YouTube author channel. Each weighted author-video edge counts that author's comments on a selected video. PageRank measures position in this participation network; it is not proof that an author influences other people. Higher edge weights increase transition strength (igraph, n.d.-a).

Readable usernames below were recovered from the saved raw authorDisplayName field and matched using channel IDs. IDs remain the analysis key because names can change. Relabelling does not alter the rankings or scores.

Artist	Unique actors	Video nodes	Sum of edge weights
Taylor Swift	2657	9	3600
Ariana Grande	2932	9	3600
Billie Eilish	3174	9	3600

Table 5. Actor-network size

Artist	Rank	Username	PageRank	Comments	Videos
Ariana Grande	1	@JfkfkfIrororkr	0.010267	107	6
Ariana Grande	2	@azizam-i3b	0.001923	18	4
Ariana Grande	3	@EduardooTarangoo	0.001702	15	3
Ariana Grande	4	@WeAreAnonymous1105	0.001359	12	1
Ariana Grande	5	@KimberleyPoole-y9n	0.001339	11	1
Billie Eilish	1	@LuismartinezFloresm	0.001258	11	1
Billie Eilish	2	@dwsunterrr6553	0.001252	11	2
Billie Eilish	3	@QuanThaSav	0.001148	10	1
Billie Eilish	4	@GabrielLunaLopez-lv5wn	0.001120	10	1
Billie Eilish	5	@user-yh3lr1gw4y	0.001037	9	1
Taylor Swift	1	@gods_angel.__x	0.006142	64	6
Taylor Swift	2	@Lexieleftwich-k8r	0.004993	59	1
Taylor Swift	3	@Christina-y5s	0.004023	37	5
Taylor Swift	4	@NevenVengdasalam-w8r	0.003356	31	3
Taylor Swift	5	@ElizabethGonzalez-ce3qn	0.002122	23	3

Table 6. Top five actors for each artist, with original results retained

Ariana's @JfkfkfIrororkr made 107 comments across six videos (2.97% of her sample), compared with Taylor's @gods_angel.__x at 64 across six (1.78%). Billie's leading actor made 11 (0.31%). These figures show different concentrations of activity, not different levels of fan loyalty.

Taylor's @Lexieleftwich-k8r ranked second despite commenting on only one video. Repetition on a single video can therefore contribute to high PageRank. Compare ranks and participation within each graph; absolute scores also depend on graph size and structure.

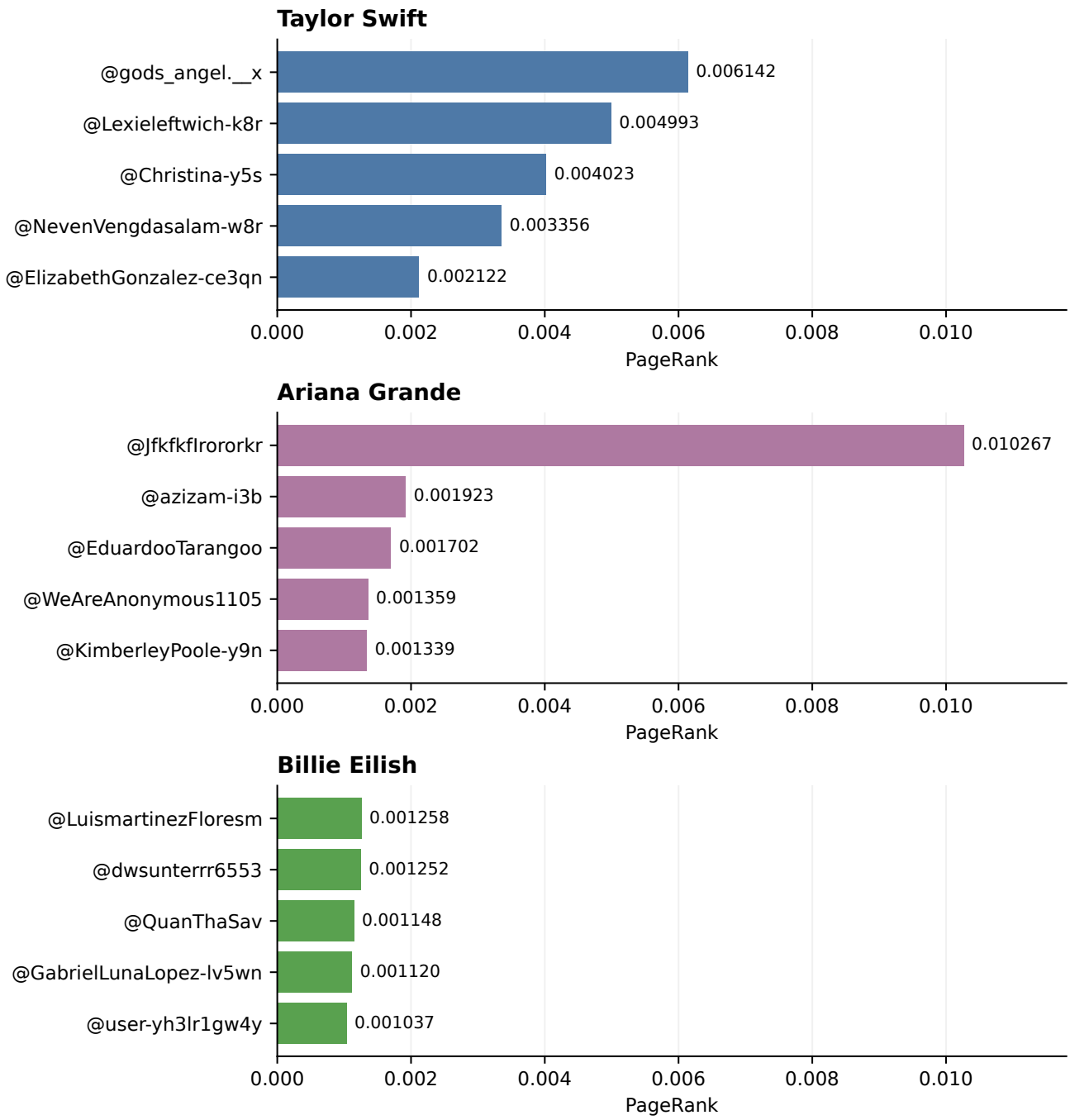


Figure 2. The original top-five PageRank results, relabelled with saved usernames.

The shared horizontal scale makes the concentration visible. Within Ariana's sample, the first actor is far ahead of her other four leading actors. Billie's leading scores are close together. Activity on these selected videos should not be generalised to either artist's whole audience.

4. Unique Actors

Artist	Comments	Unique actors	Single-comment	Repeat actors	Repeat %	Comments per actor
Taylor Swift	3600	2657	2326	331	12.46%	1.35
Ariana Grande	3600	2932	2649	283	9.65%	1.23
Billie Eilish	3600	3174	2902	272	8.57%	1.13

Table 7. Unique and repeat actors

Measure	Result
Unique actors across all datasets	8713
Actors in one artist dataset	8666
Actors in two artist datasets	44
Actors in all three datasets	3

Table 8. Cross-artist participation

Billie Eilish had the most unique actors (3,174), so her sample had the broadest participation. Taylor Swift had fewer unique actors but the highest repeat-actor percentage (12.46%) and the highest comments per actor (1.35). This suggests that Taylor's sampled audience was more concentrated and more likely to comment more than once.

Only 47 of 8,713 actors (0.54%) appeared in more than one artist sample. This is low observed overlap in these capped video samples; it does not establish that the wider fan communities are separate.

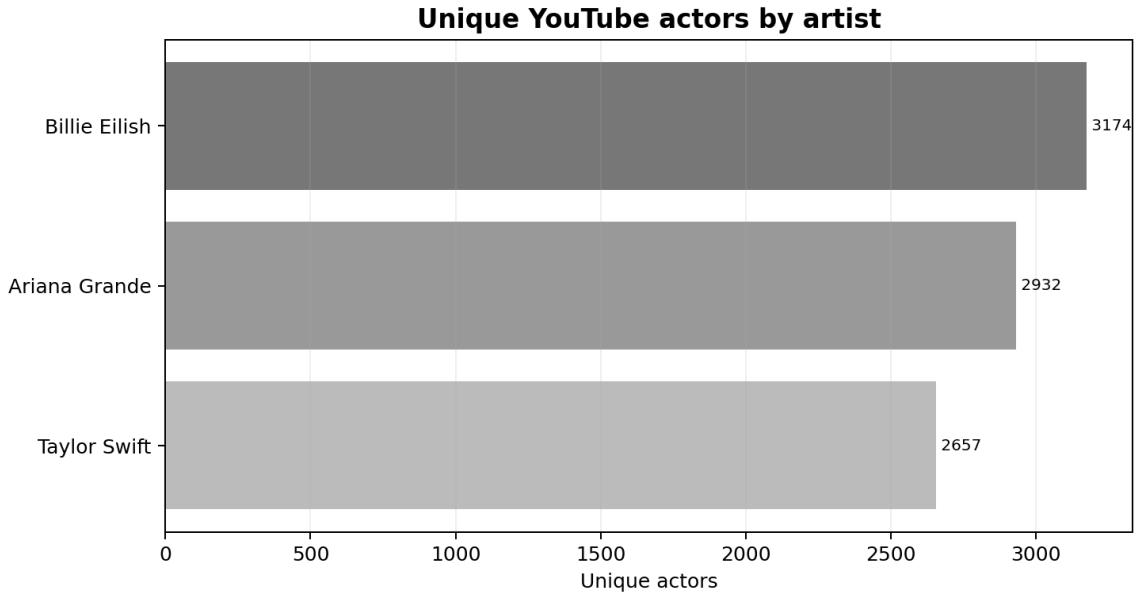


Figure 3. Unique actors in the three datasets

Counting logic in R. For one artist, let `ids` contain its `author_channel_id` values. Equivalent expressions are `counts <- table(ids)`, `length(counts)` for unique actors, `sum(counts == 1)` for single-comment actors, and `sum(counts > 1)` for repeat actors. Repeat percentage is `100 * sum(counts > 1) / length(counts)`; Taylor's 331 / 2657 gives 12.46%. IDs identify actors; usernames label them. Repeat activity can occur on one video and does not necessarily mean cross-video participation.

5. Term-Document Matrices and Top Ten Terms

One comment is one document. Lower-casing and removal of URLs, punctuation, numbers, English stop words, artist names, platform terms and selected filler produced one term-document matrix per artist. Rows represent terms, columns represent comments, and each cell counts a term in that comment. Summing a term row gives its frequency.

Artist	Documents	Unique terms	Occurrences	Empty after cleaning
Taylor Swift	3600	5756	16840	489
Ariana Grande	3600	6412	24876	452
Billie Eilish	3600	5506	18987	657

Table 9. Original term-document matrix results

The interpretations below come from reading matching comments. They describe observed contexts, not a classification of every occurrence. Frequencies and preprocessing are unchanged.

Taylor Swift

Rank	Term	Frequency	Interpretation in the sampled comments
1	love	451	Affection for Taylor and her music, but also lyrics and relationship stories; repeated uses are not separate supporters.
2	life	90	Links songs to personal experiences; also occurs in lyrics and unrelated narratives, so it has several meanings.
3	best	89	Comparative praise for songs, performances and Taylor; multiple superlatives can come from one commenter.
4	thank	81	Gratitude and replies to users; repeated messages requesting private-page follows also contribute, so sincerity is uncertain.
5	time	74	Repeat watching, first-listen memories and all-time comparisons; combines listening habits with ordinary time references.
6	happy	73	Enjoyment and good wishes, including holidays, relationships and viewing milestones; not always the song's emotional effect.
7	never	73	Occurs in Elizabeth Taylor lyrics, discovery comments and opinions; its context determines whether the message is negative.
8	beautiful	72	Praises music, visuals and appearance; some comments concern actress Elizabeth Taylor rather than Swift.
9	ever	61	Intensifies praise and criticism and occurs in lyrics; a comparison modifier rather than an independent theme.
10	elizabeth	59	Refers to the selected song, its lyrics and the actress; prominence reflects the video's subject.

Ariana Grande

Rank	Term	Frequency	Interpretation in the sampled comments
1	love	533	Combines praise, relationships and copied lyrics; repeated lyrical uses cannot be counted as separate positive opinions.
2	mind	174	Recurs in Supernatural lyrics and references to Eternal Sunshine of the Spotless Mind and the video story.
3	supernatural	160	Names the selected song and repeated chorus; also appears in production praise and video discussion.
4	petal	143	Names the song and album; some comments identify a name-tag clue in a video.
5	time	129	Listening memories, promises to revisit videos and personal messages; several contexts contribute to the count.
6	cause	115	Shortened because in Twilight Zone lyrics and conversational explanations; retained connective language, not a distinct subject.
7	yeah	108	Agreement and lyrical interjection, especially in Hate That I Made You Love Me; limited standalone meaning.
8	good	106	Evaluates music but also appears in memories, health wishes and criticism; the object being judged varies.
9	come	105	Invitations in repeated lyrics, return-listening comments and unrelated messages; not evidence by itself of a comeback.
10	never	100	Praise about not disappointing listeners, memories and personal narratives; not automatically a negative judgement.

5. Term-Document Matrices and Top Ten Terms (continued)

Billie Eilish

Rank	Term	Frequency	Interpretation in the sampled comments
1	love	673	Admiration, relationships and copied lyrics, including L'AMOUR DE MA VIE; repetition helps explain the high frequency.
2	time	185	Wildflower lyrics and first or repeated listening; combines song wording with audience listening habits.
3	life	171	L'AMOUR DE MA VIE lyrics and personal accounts of change, grief and relationships combine here.
4	beautiful	132	Praises voice, songs and appearance; some comments instead describe a scene in Barbie.
5	best	123	Performance praise, good wishes, award categories and lyrics; broader than a count of listeners preferring Billie.
6	never	110	Lyrics, emotional reactions and unfulfilled hopes; negation does not automatically mean criticism of Billie.
7	til	109	Shortened until, mainly in copied Birds of a Feather lyrics; frequent lyric wording rather than a separate theme.
8	better	107	Comparisons of live recordings or covers and copied lyrics; some comparisons favour other versions.
9	day	107	Birds of a Feather lyrics and daily-listening comments; combines song content with reported engagement.
10	back	100	Wildflower lyrics about lingering memories, return-listening comments and playback discussion; several meanings share this token.

Comparison and preprocessing limits

Love leads all three lists, but occurrences include both appreciation and quoted lyrics. Taylor's list combines evaluation and gratitude with the song reference elizabeth. Ariana's supernatural, petal and mind reflect specific releases and lyric discussion. Billie's life, time, day and til include repeated lyric language as well as personal reactions. The comment-level audit on page 11 shows why frequency is not a count of positive fans.

Words such as cause, yeah and til survive the selected stop-word list. Their lyric context is analytically useful, but they are weaker standalone themes than song names or descriptive terms. Removing them would change the matrices and rankings; the present interpretation therefore identifies this limitation rather than claiming perfectly noise-free text. Empty comments remain in the Q5 matrices as zero columns; later models that require non-empty text use the smaller subsets reported in Q11.

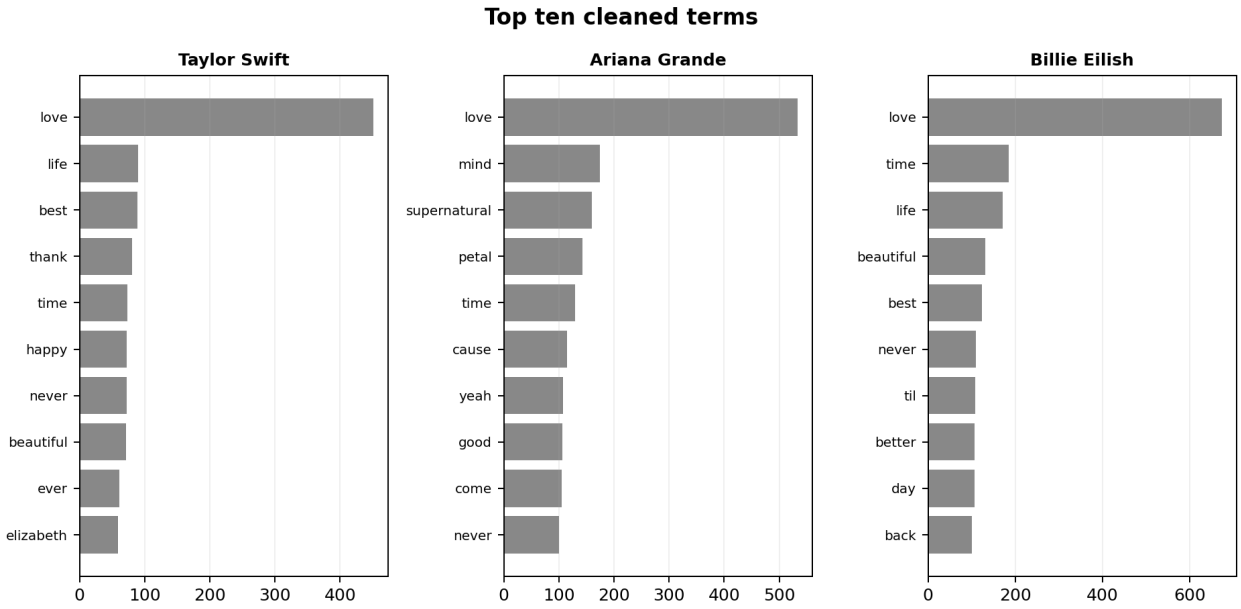


Figure 4. Original top-ten term frequencies; counts and rankings unchanged.

5.1 Where does the high count of love come from?

The final 10,800-comment file was checked without changing its records. The audit lowercases text, decodes HTML entities, removes URLs, ASCII punctuation and digits, then counts the whole retained token love. This reproduces the reported frequencies exactly; loves and loved are not counted as love.

Artist	Occurrences	Comments with love	% of 3,600 comments	Distinct authors	Extra uses within comments
Taylor Swift	451	384	10.67%	315	67
Ariana Grande	533	360	10.00%	336	173
Billie Eilish	673	464	12.89%	448	209

Table 9a. Occurrences, documents and actors are different counts

Billie has the highest count both for occurrences (673) and comments containing the word (464). However, Ariana has more occurrences than Taylor (533 versus 451) but fewer comments containing it (360 versus 384). Repetition within comments reverses their ordering. The extra-use column counts uses beyond the first in each comment; it does not identify all repeated or copied comments.

Artist	Three leading source videos per artist	Uses of love	Share of artist total
Taylor Swift	Elizabeth Taylor - music video	90	19.96%
Taylor Swift	I Can Do It With a Broken Heart - official video	68	15.08%
Taylor Swift	Fortnight - music video	53	11.75%
Ariana Grande	we can't be friends - music video	104	19.51%
Ariana Grande	hate that i made you love me - lyric video	90	16.89%
Ariana Grande	supernatural - music video	77	14.45%
Billie Eilish	L'AMOUR DE MA VIE - Songline	156	23.18%
Billie Eilish	WILDFLOWER - FireAid	92	13.67%
Billie Eilish	WILDFLOWER - COLORS	90	13.37%

Table 9b. Video sources of the original love frequencies

Lyric copying. Three comments under Ariana's we can't be friends reproduce the lyrics and contain 15 uses each. Together they contribute 45 of 533 occurrences (8.44%), from just three comments. Six lyric-copy comments under Billie's L'AMOUR DE MA VIE Songline performance contain nine uses each: 54 of 673 (8.02%). These are inspected examples, not a complete classification of lyric comments.

Praise and other contexts. Direct appreciation also appears, for example “I love you Ari” and “Love her voice. Queen”. Other comments discuss relationships or quote song wording. Taylor's largest count within one comment is four, compared with 15 for Ariana and nine for Billie.

Interpretation. The word reflects a mixture of appreciation, lyrics and personal discussion. Repeated uses cannot be counted as separate positive opinions. No lyrics or other records were removed for this audit, so the original term, PageRank, sentiment and topic results remain intact.

6. Semantic Networks and PageRank

A directed network links consecutive terms in each cleaned comment; edge weights count retained bigram occurrences. Stop-word removal can join words that were not adjacent in the original sentence. The graph therefore describes transitions in processed text, not full meaning or direct sentiment. Short fragments were removed before the final ranking.

Taylor Swift

Rank	Term	PageRank
1	love	0.026013
2	ever	0.007950
3	feel	0.007467
4	forever	0.006703
5	elizabeth	0.006525
6	worthless	0.006162
7	mind	0.006111
8	life	0.006036
9	place	0.006009
10	hope	0.005586

Ariana Grande

Rank	Term	PageRank
1	love	0.013498
2	david	0.010956
3	god	0.009230
4	cause	0.006920
5	supernatural	0.006545
6	time	0.006410
7	mind	0.006023
8	verse	0.006009
9	yeah	0.005648
10	abraham	0.005589

Billie Eilish

Rank	Term	PageRank
1	love	0.027101
2	time	0.015419
3	til	0.014418
4	life	0.014306

Rank	Term	PageRank
5	better	0.009407
6	day	0.009246
7	eyes	0.009170
8	said	0.007983
9	never	0.007854
10	mind	0.007774

Artist	Shared top-ten terms	Shared terms between frequency and PageRank
Taylor Swift	4	love, life, ever, elizabeth
Ariana Grande	6	love, mind, supernatural, time, cause, yeah
Billie Eilish	7	love, time, life, never, til, better, day

Table 10. Agreement between frequency and PageRank rankings

Artist	Strong bigram	Weight
Taylor Swift	high school	22
Taylor Swift	thank love	18
Taylor Swift	feel worthless	17
Taylor Swift	fate ophelia	16
Taylor Swift	lana del	16
Ariana Grande	wait love	60
Ariana Grande	hate love	57
Ariana Grande	supernatural loves	52
Ariana Grande	believe happened	49
Ariana Grande	possessin mind	46
Billie Eilish	love life	63
Billie Eilish	day die	50
Billie Eilish	til day	50
Billie Eilish	back mind	49
Billie Eilish	fall love	43

Table 11. Five strongest bigram connections per artist

Love ranks first within all three semantic networks. Frequent lyric sequences can strengthen its links: Billie's love-life and til-day pairs and Ariana's wait-love pair fit the lyric-copy comments examined in the audit. High PageRank therefore combines network structure with repetition; it does not independently establish admiration.

Frequency and PageRank share four top-ten terms for Taylor, six for Ariana and seven for Billie. Frequency counts tokens; weighted PageRank also depends on the direction and strength of incoming links (igraph, n.d.-a). A lower-frequency term can rank highly inside a repeated, closely linked passage. Absolute scores across different graphs should be compared cautiously.

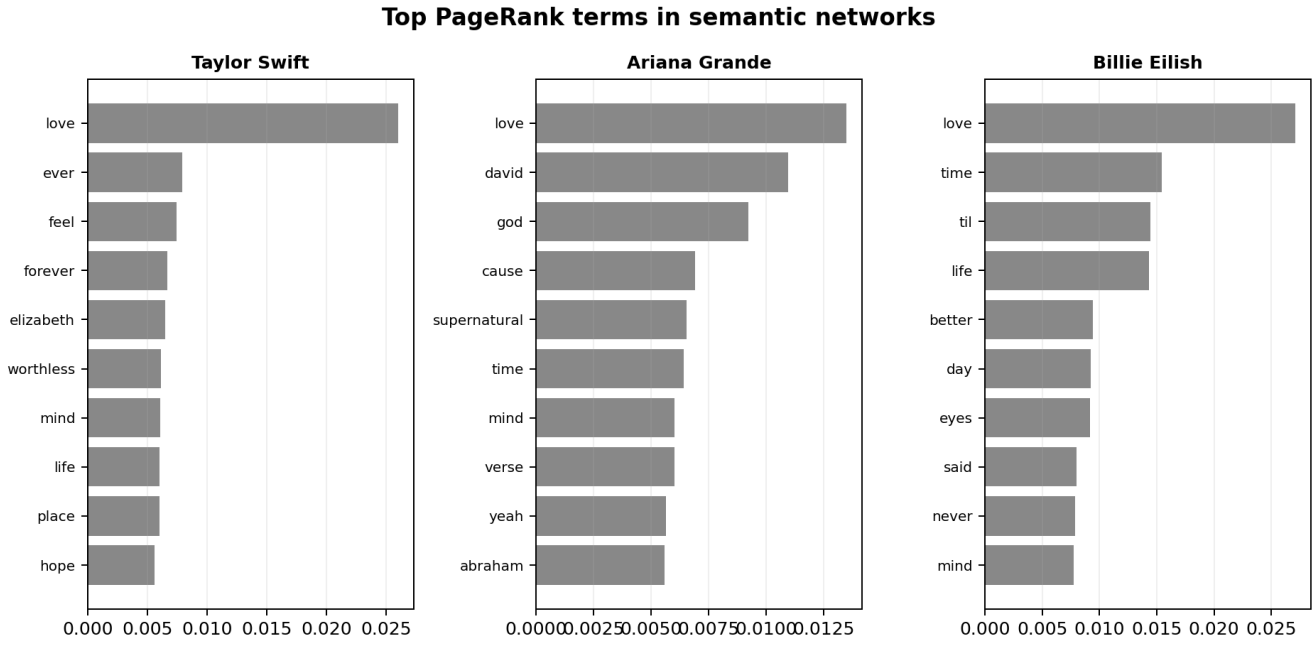


Figure 5. Original semantic PageRank rankings; values unchanged.

Checking unexpected high-ranking words

Ariana's david and abraham are not evidence of music-related themes. Checking the final comment file found 58 uses of david in six comments. Two identical 4,134-character religious posts contributed 27 uses each: 54 of 58 (93.10%). Abraham appeared 24 times, all within two identical 3,666-character religious posts.

Those posts have different comment IDs, so removing duplicate IDs did not remove their repeated text. Their recurring word sequences provide a concrete explanation for the prominence of these names in PageRank and in Ariana's first LDA topic. The ranking is sensitive to repeated off-topic passages as well as musical discussion.

The reported scores are retained. These comments have not been deleted retrospectively, and no claim is made about a recalculated ranking without them. This check narrows the interpretation: the graph captures the language present in the sampled comments, including off-topic material.

Term	Occurrences	Comments	Source of concentration
david	58	6	54 uses in two identical long posts
abraham	24	2	All 24 uses in two identical long posts

Source: contextual audit of the final Comments.csv; all original records retained.

6. Frequency and PageRank: Direct Comparison

Figure 5a compares rank positions under the two methods for each artist. Rank 1 is highest; the reversed horizontal scale makes the longest bar the highest-ranked term. These bars show ranks, not raw token counts or PageRank scores.

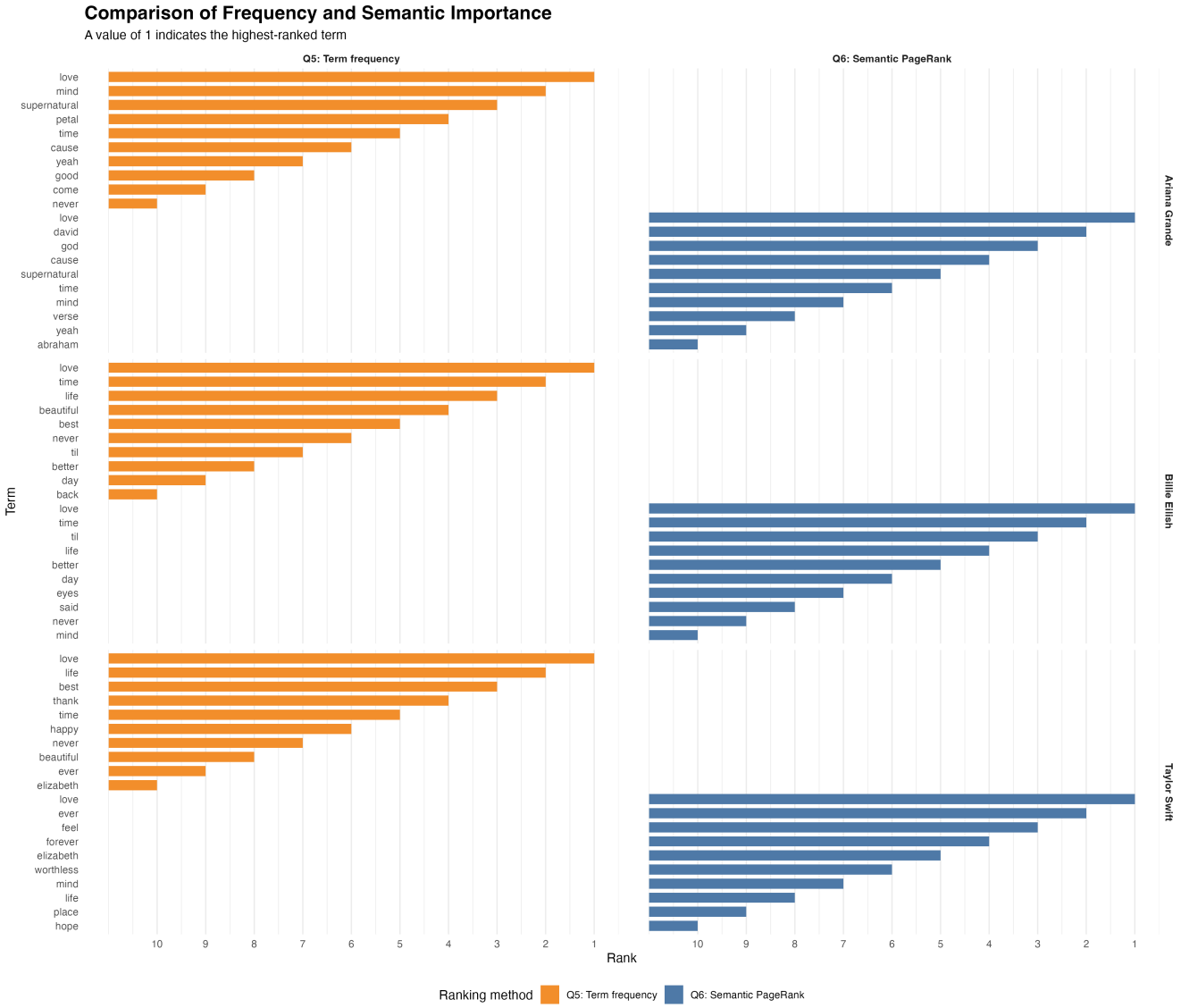


Figure 5a. Existing R comparison of the Q5 frequency top ten and the Q6 weighted PageRank top ten.

Taylor: only four terms appear in both top tens. Ever rises from frequency rank 9 to PageRank rank 2, and elizabeth from 10 to 5, while life falls from 2 to 8. Frequently repeated words and words occupying influential positions in the retained bigram graph are therefore not interchangeable.

Ariana: six terms overlap. Mind falls from 2 to 7, while cause rises from 6 to 4. Petal is fourth by frequency but outside the PageRank top ten. David and abraham enter the PageRank top ten; the preceding comment audit traces their prominence to repeated off-topic passages.

Billie: seven terms overlap, the highest agreement. Love and time remain first and second, while til rises from 7 to 3 and better from 8 to 5. Repeated lyric sequences such as til-day and love-life help explain the agreement and shifts. A term absent from one top-ten list is not necessarily absent from that corpus or graph.

Together, frequency identifies repeated vocabulary, while PageRank highlights structure in the retained transitions. Both are affected by repeated text; neither alone measures audience approval. Source: Q6_Frequency_PageRank_Comparison.csv and the existing Q6 plot export.

7. Centrality Analysis

Degree counts distinct videos and weighted degree counts comments. For the reported path scores, comment weights are converted to distances as 1/weight: heavier participation gives a shorter edge. Betweenness measures shortest-path brokerage; normalised closeness is inverse average path distance (igraph, n.d.-b, n.d.-c).

Artist	Vertices	Edges	Actors	Max degree	Max weighted degree	Max betweenness	Max closeness
Taylor Swift	2666	2813	2657	6	64	0.3551	0.7612
Ariana Grande	2941	3061	2932	6	107	0.5117	0.6704
Billie Eilish	3183	3246	3174	6	11	0.1524	0.4178

Table 12. Network and maximum centrality results

Artist	Mean degree	Mean weighted degree	Mean betweenness	Mean closeness
Taylor Swift	1.0587	1.3549	0.000557	0.4272
Ariana Grande	1.0440	1.2278	0.000467	0.3749
Billie Eilish	1.0227	1.1342	0.000332	0.2751

Table 13. Average centrality comparison

Ariana's @JfkfkfIrororkr has the highest recorded betweenness (0.5117) and weighted degree (107). This actor connects six videos and is a major bridge in the sampled participation graph. Taylor has the highest mean degree, weighted degree, betweenness and closeness, indicating more cross-video participation on average under this graph definition.

Breadth and repetition are different. Taylor's @Lexieleftwich-k8r made 59 comments on one video, while Billie's @ye2e made six comments across six videos. The first actor is highly active in one location; the second links a wider set of videos. Neither pattern alone proves social influence.

The printed component check confirms one connected component for each artist. Connectivity requires cross-video authors to link the video groups; simply giving each author a video edge would not guarantee it. These graphs link authors to videos, not authors replying to one another.

Table 12 reports raw maximum actor degree; the degree rankings in Figure 6 use degree/(all graph vertices - 1). That denominator includes authors and videos. Since an author can connect to only nine selected videos, small normalised degree values should not be interpreted as low interest. A raw degree of six means participation on six of nine videos.

All comparisons are conditional on the capped sample, video selection and inverse-weight distance convention. They describe observed participation and modelled paths, not actual information flow, friendship or the whole fan population.

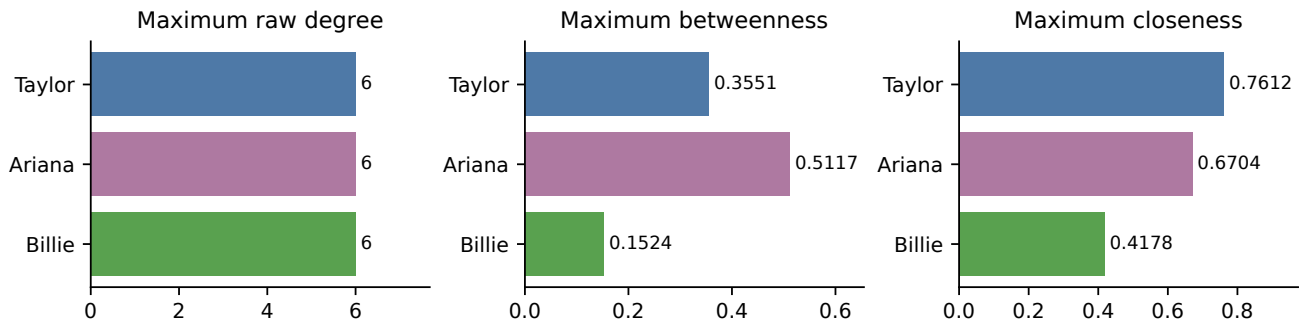


Figure 6. Centrality comparison and top-five actor rankings

Taylor Swift

Rank	Degree: username	Score	Betweenness: username	Score	Closeness: username	Score
1	@gods_angel.__x	0.002251	@gods_angel.__x	0.355141	@Christina-y5s	0.761152
2	@crimsoncl9ver	0.002251	@AlirezaPakipersianrap	0.227808	@gods_angel.__x	0.759351
3	@WarrenYow	0.002251	@ElizabethGonzalez-ce3qn	0.193236	@NevenVengdasalam-w8r	0.748713
4	@Christina-y5s	0.001876	@Christina-y5s	0.169958	@Lexieleftwich-k8r	0.746380
5	@teapee6363	0.001501	@ec2official	0.145060	@AlirezaPakipersianrap	0.733869

Ariana Grande

Rank	Degree: username	Score	Betweenness: username	Score	Closeness: username	Score
1	@JfkfkfIrororkr	0.002041	@JfkfkfIrororkr	0.511657	@JfkfkfIrororkr	0.670357
2	@maylennaasley	0.001701	@EduardooTarangoo	0.224050	@EduardooTarangoo	0.628577
3	@jessicawilliams1176	0.001701	@HazielTarango	0.143356	@azizam-i3b	0.606008
4	@yesitiswinniethepooh	0.001701	@shawniaconley5980	0.102714	@doney1996	0.584228
5	@AirRusher1992	0.001701	@azizam-i3b	0.095322	@TheStoryTeller-z3b	0.584228

Billie Eilish

Rank	Degree: username	Score	Betweenness: username	Score	Closeness: username	Score
1	@ye2e	0.001886	@Kipresentes	0.152356	@ye2e	0.417784
2	@Kipresentes	0.001571	@ye2e	0.091306	@dwsunterrr6553	0.408331
3	@FIXA1-9	0.000943	@Echoesspram9	0.084018	@Kipresentes	0.408305
4	@Echoesspram9	0.000943	@ThatRedhed	0.074329	@user-yh3lr1gw4y	0.400261
5	@deonteterrel8078	0.000943	@dwsunterrr6553	0.071949	@PatrigPoulaouen	0.393457

Names are matched to saved author IDs. Scores and the original tie order are retained. Degree scores in the ranking tables are normalised; the upper chart shows raw maximum degree. Betweenness and closeness use inverse comment-weight distances.

8. Community Analysis

Girvan-Newman is the required method for Q8. Louvain is included as an additional comparison, not a replacement. Both methods were applied to author-video networks restricted to authors who commented on at least two selected videos. This focuses on cross-video participation and reduces computation; the partitions describe this active subset, not the full audience.

Artist	Algorithm	Active actors	Videos	Edges	Communities	Modularity	Largest	Largest %
Taylor Swift	Girvan-Newman	110	9	266	7	0.1617	33	27.73%
Taylor Swift	Louvain	110	9	266	5	0.4794	27	22.69%
Ariana Grande	Girvan-Newman	90	9	219	4	0.1720	34	34.34%
Ariana Grande	Louvain	90	9	219	5	0.4920	34	34.34%
Billie Eilish	Girvan-Newman	58	9	130	5	0.2798	20	29.85%
Billie Eilish	Louvain	58	9	130	5	0.4494	18	26.87%

Table 14. Community detection results

Artist	Adjusted Rand agreement between algorithms
Taylor Swift	0.1811
Ariana Grande	0.2295
Billie Eilish	0.1957

Table 15. Agreement between Girvan-Newman and Louvain

For the reported partitions, Louvain has higher weighted modularity for all three artists: Ariana 0.4920, Taylor 0.4794 and Billie 0.4494. Within each graph, these partitions have stronger separation by this criterion than the reported Girvan-Newman partitions. Higher modularity does not establish a more socially cohesive or loyal fandom.

The retained actors represent only 4.14% of Taylor's sampled authors (110/2,657), 3.07% of Ariana's (90/2,932), and 1.83% of Billie's (58/3,174). Most people commented on one selected video and are absent from this analysis. Their exclusion changes the question from the full audience to the small cross-video subset.

Adjusted Rand agreement is low: 0.1811, 0.2295 and 0.1957. Even when both methods return five communities for Billie, the same count does not mean the same memberships. Community numbers are arbitrary labels, not shared groups across artists or algorithms.

Girvan-Newman removes edges with high edge betweenness; Louvain seeks a high-modularity partition. Edge betweenness treats weights as distances, whereas Louvain treats them as connection strengths (igraph, n.d.-d, n.d.-e). The different objectives and weight conventions matter when comparing outputs.

These are groups of authors and videos with related participation patterns. The edges do not record friendships or direct replies. Without a sensitivity check using another selection or graph definition, the partitions should be treated as exploratory rather than stable social groups.

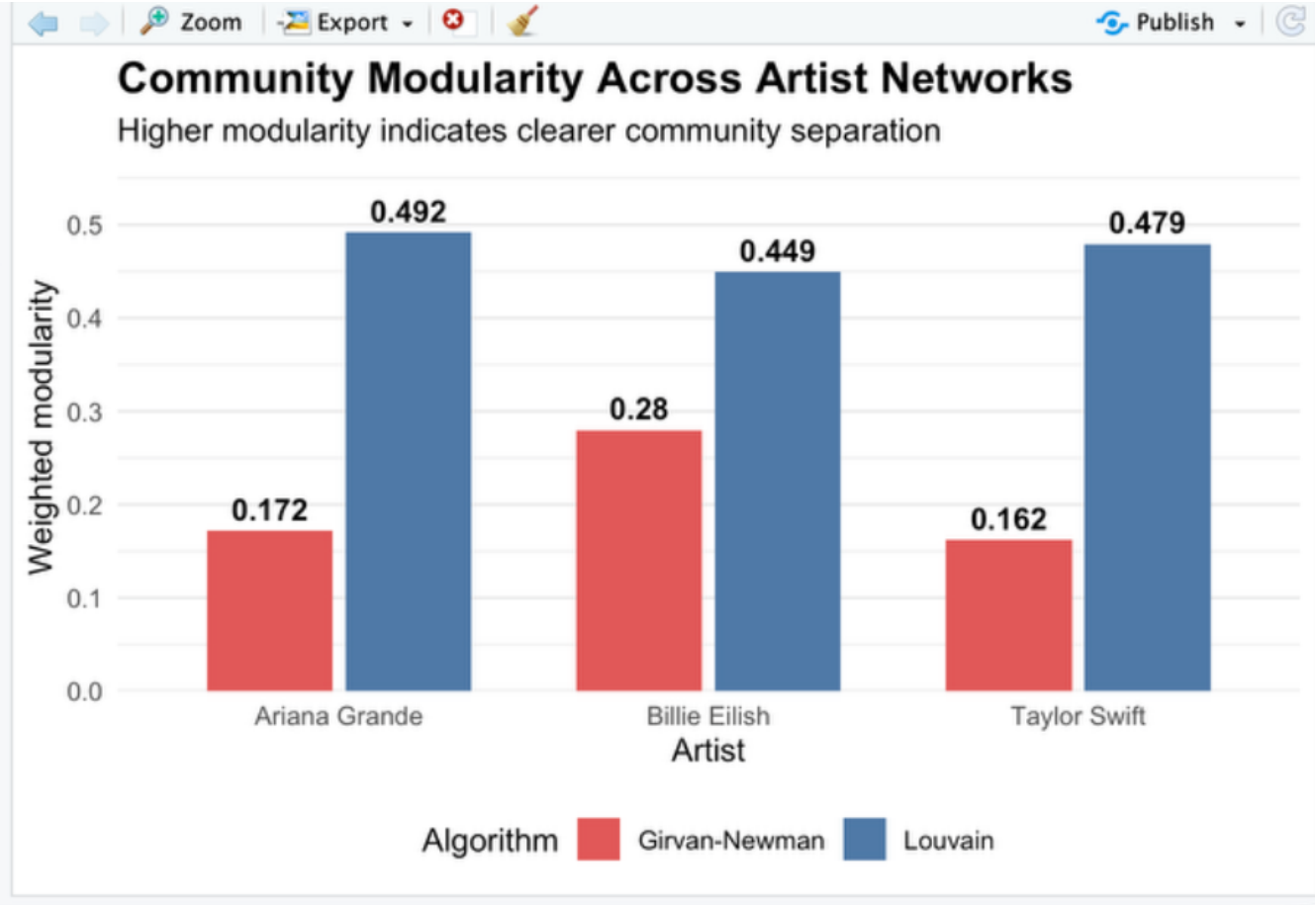


Figure 7. Original community-modularity output, cropped to the plot.

```
  "03_Clean_Data/Q8_Video_Community_Comparison.csv",
  row.names = FALSE,
  fileEncoding = "UTF-8"
)

save(
  q8_taylor,
  q8_ariana,
  q8_billie,
  q8_community_summary,
  q8_membership_table,
  q8_community_composition,
  q8_video_communities,
  q8_community_count_plot,
  q8_modularity_plot,
  file = "04_RData/Q8_Community_Analysis.RData"
)

print(q8_community_summary)

print(q8_video_communities)

print(q8_modularity_plot)
```

Supporting source-code screenshot from the same RStudio run.

8. Community Counts and Method Comparison

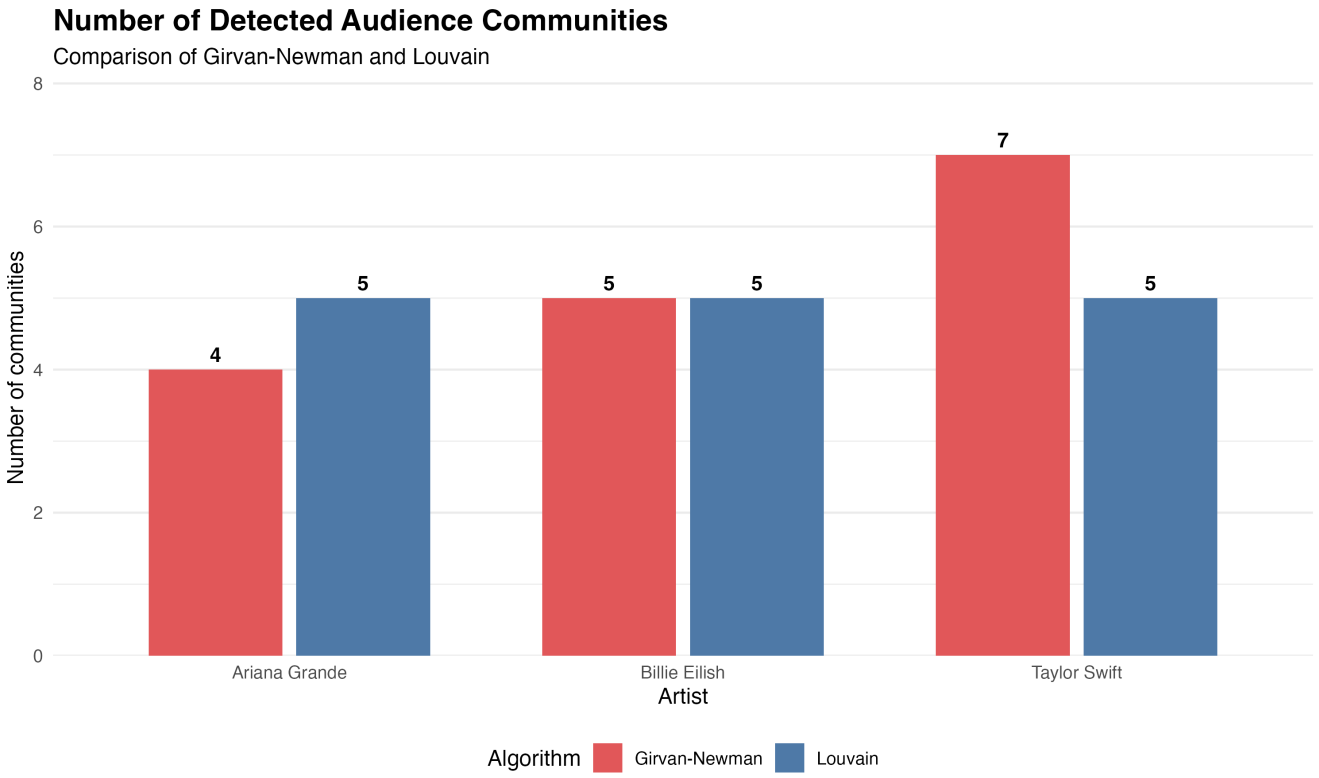


Figure 7a. Community counts in the same restricted cross-video author-video networks under both algorithms.

Girvan-Newman returns seven groups for Taylor, four for Ariana and five for Billie. Louvain returns five for every artist. The count difference is largest for Taylor, but a larger count is not automatically a better partition. For Billie, identical counts coexist with adjusted Rand agreement of only 0.1957: five communities can contain different members.

How the network figures should be read

The following paired plots retain identical node positions and edges within each artist. Circles are authors who commented on at least two selected videos; squares are videos, labelled V1-V9. Colour represents algorithm membership. Community numbers and colours are local to each panel, so a colour must not be treated as the same group in another panel. Node size indicates type, not centrality.

Edge weights are author comment counts on a video. Girvan-Newman uses reciprocal counts (1 / weight) as path distances; Louvain uses counts as connection strengths. Reported modularity uses the original count weights for both partitions. Displayed edge widths are logarithmically scaled, rather than proportional to raw counts.

The algorithms answer the same exploratory grouping question through different procedures. Louvain's higher modularity is consistent with its optimisation objective; it is not evidence that its communities are real friendship groups. The plots concern 110, 90 and 58 cross-video authors for Taylor, Ariana and Billie, respectively, not every commenter in the full samples.

Sources: Q8_Community_Analysis_Summary.csv, Q8_Community_Memberships.csv and the existing Q8 network exports. The V labels below follow the plotted video-node order.

8. Community Partitions: Taylor Swift

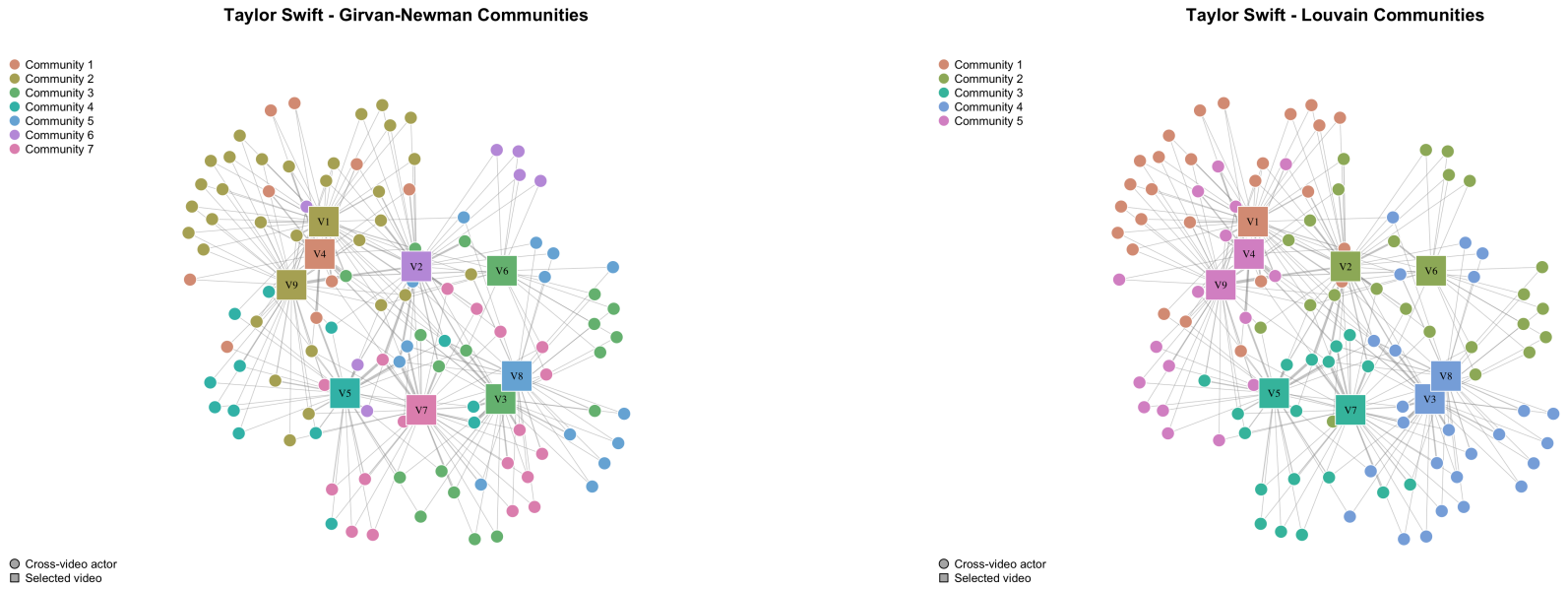


Figure 7b. Girvan-Newman (left) and Louvain (right), with fixed positions within the pair. Circles are cross-video authors; squares are videos. Membership colours are specific to each panel.

Girvan-Newman divides 119 nodes into seven communities, compared with five for Louvain. Opalite (V1) and Elizabeth Taylor (V9) share a Girvan-Newman group but separate under Louvain. Louvain instead groups Elizabeth Taylor with The Fate of Ophelia music video (V4). Weighted modularity is 0.1617 versus 0.4794, with adjusted Rand agreement of 0.1811. The example shows a change in membership, not simply fewer groups.

Video key

V1 Opalite (music video)

V2 Opalite (lyric video)

V3 I Hate It Here (lyric video)

V4 The Fate of Ophelia (music video)

V5 Fortnight (music video)

V6 The Fate of Ophelia (lyric video)

V7 I Can Do It With a Broken Heart (video)

V8 So High School (lyric video)

V9 Elizabeth Taylor (music video)

8. Community Partitions: Ariana Grande

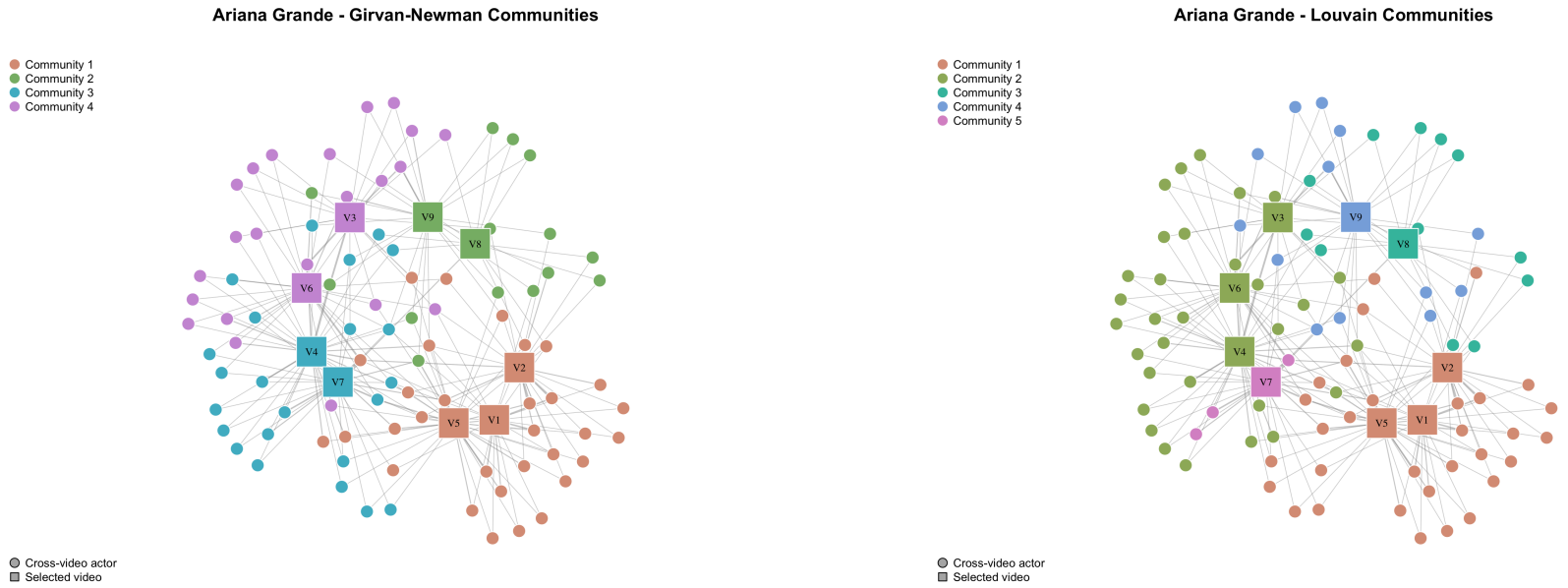


Figure 7c. Girvan-Newman (left) and Louvain (right), with fixed positions within the pair. Circles are cross-video authors; squares are videos. Membership colours are specific to each panel.

Girvan-Newman detects four communities and Louvain five among 99 nodes. Both place the two supernatural videos (V1, V5) with twilight zone (V2). Other assignments change: petal (V8) and the hate that i made you love me lyric video (V9) share a Girvan-Newman group but separate under Louvain. Weighted modularity is 0.1720 versus 0.4920; adjusted Rand agreement is 0.2295. Some video associations persist while others depend on the method.

Video key

V1 supernatural (lyric visualizer)

V2 twilight zone (lyric visualizer)

V3 hate that i made you love me (music video)

V4 yes, and? (music video)

V5 supernatural (music video)

V6 we can't be friends (music video)

V7 the boy is mine (music video)

V8 petal (lyric video)

V9 hate that i made you love me (lyric video)

8. Community Partitions: Billie Eilish

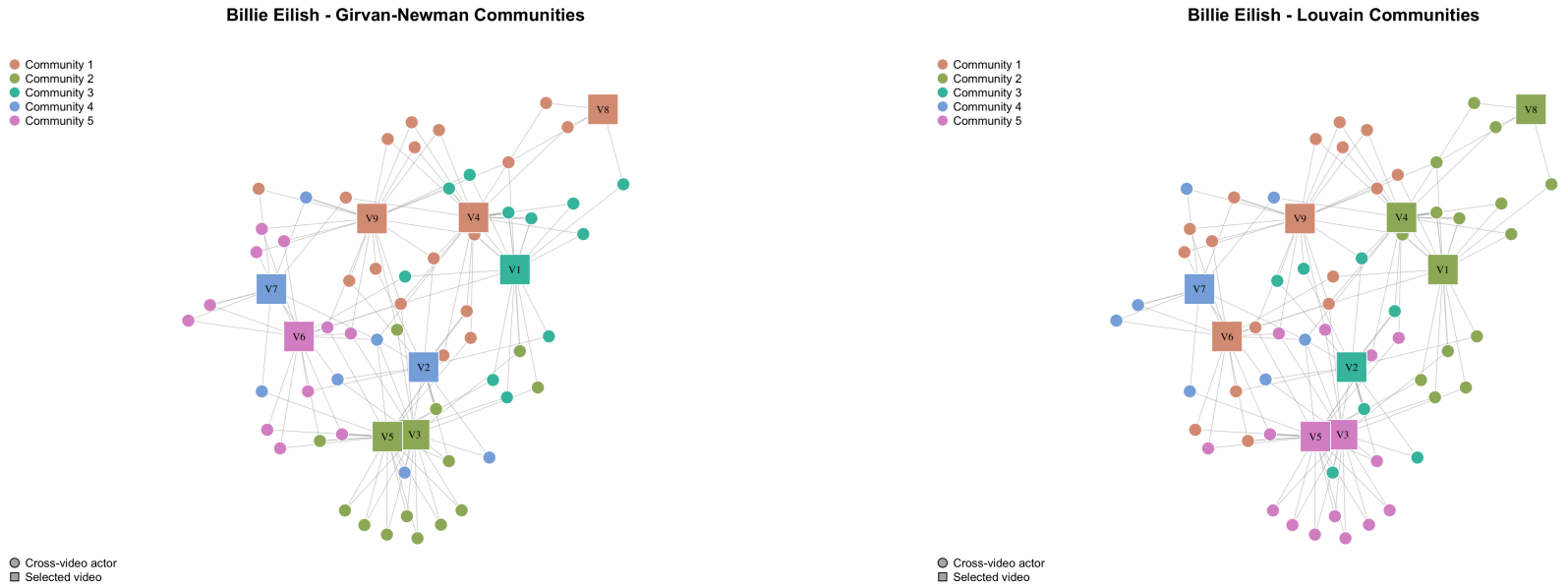


Figure 7d. Girvan-Newman (left) and Louvain (right), with fixed positions within the pair. Circles are cross-video authors; squares are videos. Membership colours are specific to each panel.

Both methods detect five communities among 67 nodes, yet adjusted Rand agreement is only 0.1957. Girvan-Newman places all three Amazon Songline performances (V4, V8, V9) together. Louvain separates BIRDS OF A FEATHER (V9) from the other two and groups it with the Olympic BIRDS OF A FEATHER performance (V6). Weighted modularity is 0.2798 versus 0.4494. This distinguishes a shared performance-series grouping from a shared-song grouping in the observed participation.

Video key

V1 WILDFLOWER (COLORS)

V2 L'AMOUR DE MA VIE (extended edit)

V3 WILDFLOWER (FireAid)

V4 L'AMOUR DE MA VIE (Songline)

V5 THE GREATEST (Olympic Handover)

V6 BIRDS OF A FEATHER (Olympic Handover)

V7 What Was I Made For? (Barbie Montage)

V8 WILDFLOWER (Songline)

V9 BIRDS OF A FEATHER (Songline)

9. Sentiment Analysis

Comment sentiment was scored and divided into positive, neutral and negative classes. The same thresholds were used for all three datasets: scores above 0.05 were positive, scores below -0.05 were negative, and the remaining scores were neutral. NRC emotion categories were also counted.

Artist	Comments	Mean	Positive	Neutral	Negative	Positive %	Neutral %	Negative %
Taylor Swift	3600	0.1183	1446	1743	411	40.17%	48.42%	11.42%
Ariana Grande	3600	0.0890	1317	1808	475	36.58%	50.22%	13.19%
Billie Eilish	3600	0.1062	1338	1862	400	37.17%	51.72%	11.11%

Table 16. Sentiment distribution

Artist	Result	Video	Mean	Positive %	Negative %
Taylor Swift	Most positive	Elizabeth Taylor (Official Music Video)	0.1977	52.50%	7.75%
Taylor Swift	Least positive	I Hate It Here (Official Lyric Video)	0.0399	32.25%	24.75%
Ariana Grande	Most positive	twilight zone (lyric visualizer)	0.1310	44.50%	9.00%
Ariana Grande	Least positive	the boy is mine (Official Music Video)	0.0221	23.75%	16.25%
Billie Eilish	Most positive	WILDFLOWER (COLORS Performance)	0.1305	38.75%	9.00%
Billie Eilish	Least positive	What Was I Made For? (Barbie Montage)	0.0723	36.00%	16.75%

Table 17. Most and least positive videos

Artist	Top emotion	Second	Third	Fourth
Taylor Swift	Joy 408.06	Trust 272.78	Anticipation 238.61	Sadness 156.67
Ariana Grande	Joy 448.89	Trust 359.17	Anticipation 300.28	Sadness 247.22
Billie Eilish	Joy 421.67	Trust 269.17	Anticipation 243.06	Sadness 220.83

Table 18. Leading NRC emotions per 1,000 comments

Taylor has the highest mean score (0.1183) and positive share (40.17%), while Ariana has the lowest mean (0.0890) and highest negative share (13.19%). These are descriptive differences in the sampled texts, not evidence that one artist is more popular. The positive-share gap between Taylor and Ariana is 3.59 percentage points.

Neutral is the largest class in each sample. It means a score between -0.05 and 0.05 under this method, not necessarily indifference: short text, unrecognised language and a mixture of positive and negative expressions can also produce a near-zero score. Lyrics about loss or conflict may score negatively without criticising the artist.

Joy is the leading NRC category. Table 18 gives emotion-word matches per 1,000 comments, not percentages of people feeling each emotion. Categories can overlap, and long comments can contribute several matches. Ariana's higher counts cannot by themselves establish a more emotionally intense audience.

The love audit and repeated off-topic posts show why context matters. Full lyric copies and long passages can affect emotion counts. Video format also differs across artists. Sentiment and topic results should therefore be read alongside the original text, rather than as direct measures of fan approval.

9. Sentiment Distribution and Source Code

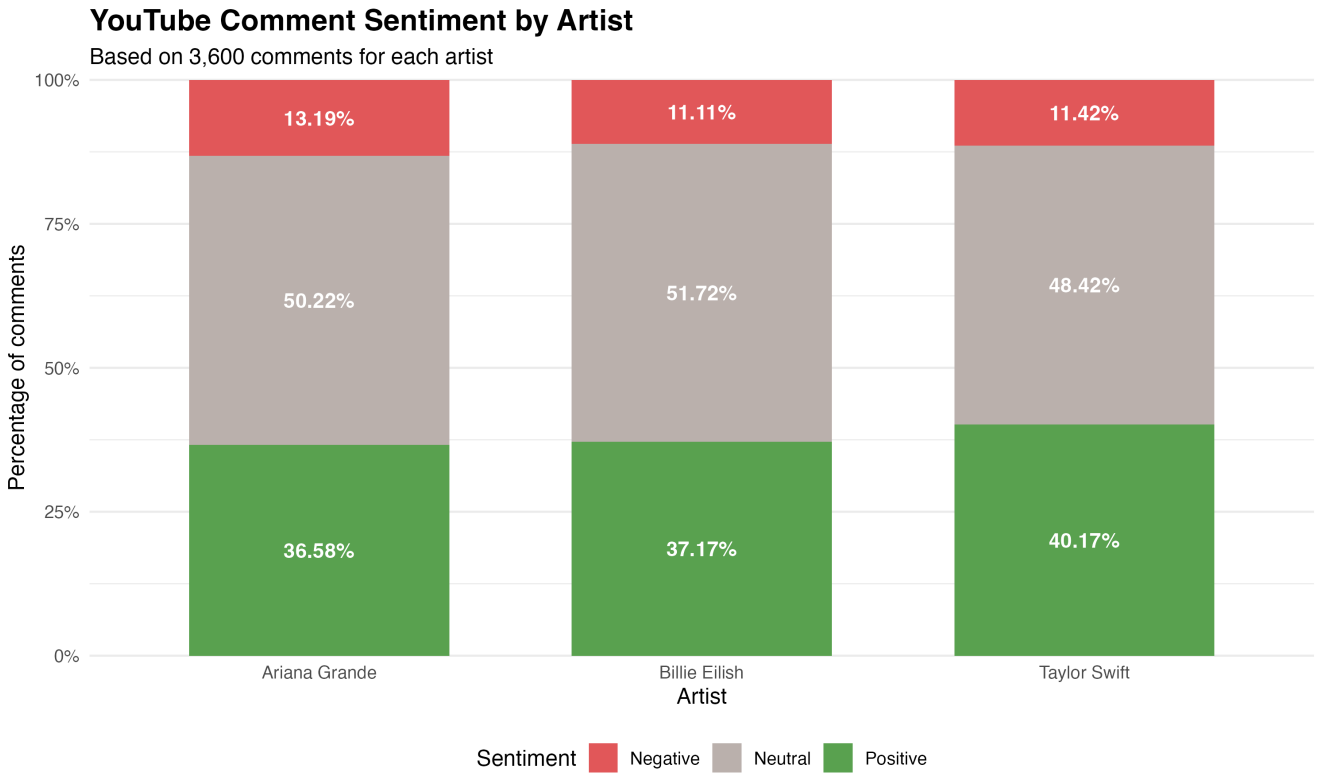


Figure 8. Saved R sentiment-distribution plot for the three artist samples.

Relevant source-code excerpt

```
q9_positive_threshold <- 0.05
q9_negative_threshold <- -0.05

q9_sentiment_data$sentiment_category <- ifelse(
  q9_sentiment_data$sentiment_score >
    q9_positive_threshold,
  "Positive",
  ifelse(
    q9_sentiment_data$sentiment_score <
      q9_negative_threshold,
    "Negative",
    "Neutral"
  )
)

print(q9_artist_summary)

print(q9_video_extremes)

print(q9_emotion_long)

print(q9_sentiment_distribution_plot)
```

Verbatim excerpt from Q9_Sentiment_Analysis.R: sentiment thresholds, classification and final display commands. This is typeset source code, not a screenshot of a new RStudio execution.

The thresholds are applied identically across artists. The displayed totals remain the saved Q9 results; this presentation update does not rerun sentiment scoring or change the sample.

9. Emotion Vocabulary Across Artists

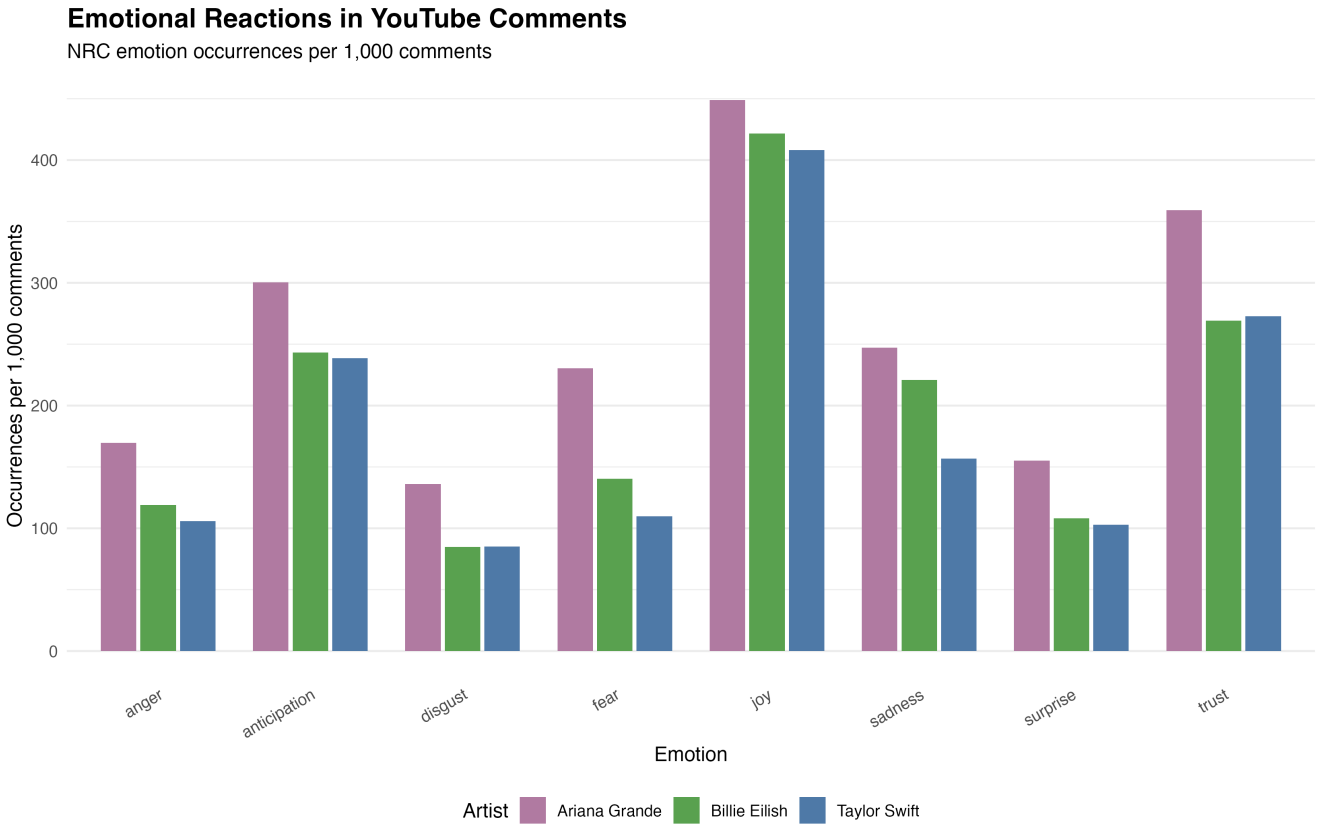


Figure 8a. NRC emotion-word matches per 1,000 sampled comments, using 3,600 comments for each artist. These are lexical match rates, not percentages of people experiencing an emotion.

Shared patterns and differences

Joy, trust, anticipation and sadness are the four leading categories for all three artists. Ariana has the highest match rate in every category, including joy (448.89), trust (359.17) and sadness (247.22). Billie follows Ariana for joy and sadness, while Taylor has slightly more trust matches than Billie (272.78 versus 269.17).

Ariana's higher emotion-match rates coexist with its lower mean sentiment score (0.0890, compared with 0.1183 for Taylor and 0.1062 for Billie). The measures describe different properties: emotion categories identify matching vocabulary, whereas sentiment gives a signed score under the scoring method. More emotion words do not necessarily imply a more positive audience.

Interpretation and limitations

The denominator controls for the equal number of comments, but not comment length. One long lyric copy can supply several matches, and one word can belong to more than one NRC category. The bars therefore need not sum to 1,000. Quoted lyrics, repeated text and off-topic passages can contribute to the counts, as the Q5/Q6 contextual checks demonstrate.

The chart is useful for selecting vocabulary and comments for closer inspection, rather than diagnosing fans' emotions. No significance test or causal claim is made from these differences. Source: the saved Q9_NRC_Emotion_Summary.csv and Q9_NRC_Emotions.png.

9. Sentiment Differences Between Videos

The artist averages hide variation between the nine selected uploads. Each bar below is a mean sentiment score from 400 sampled comments; it is not the percentage of comments classified as positive.

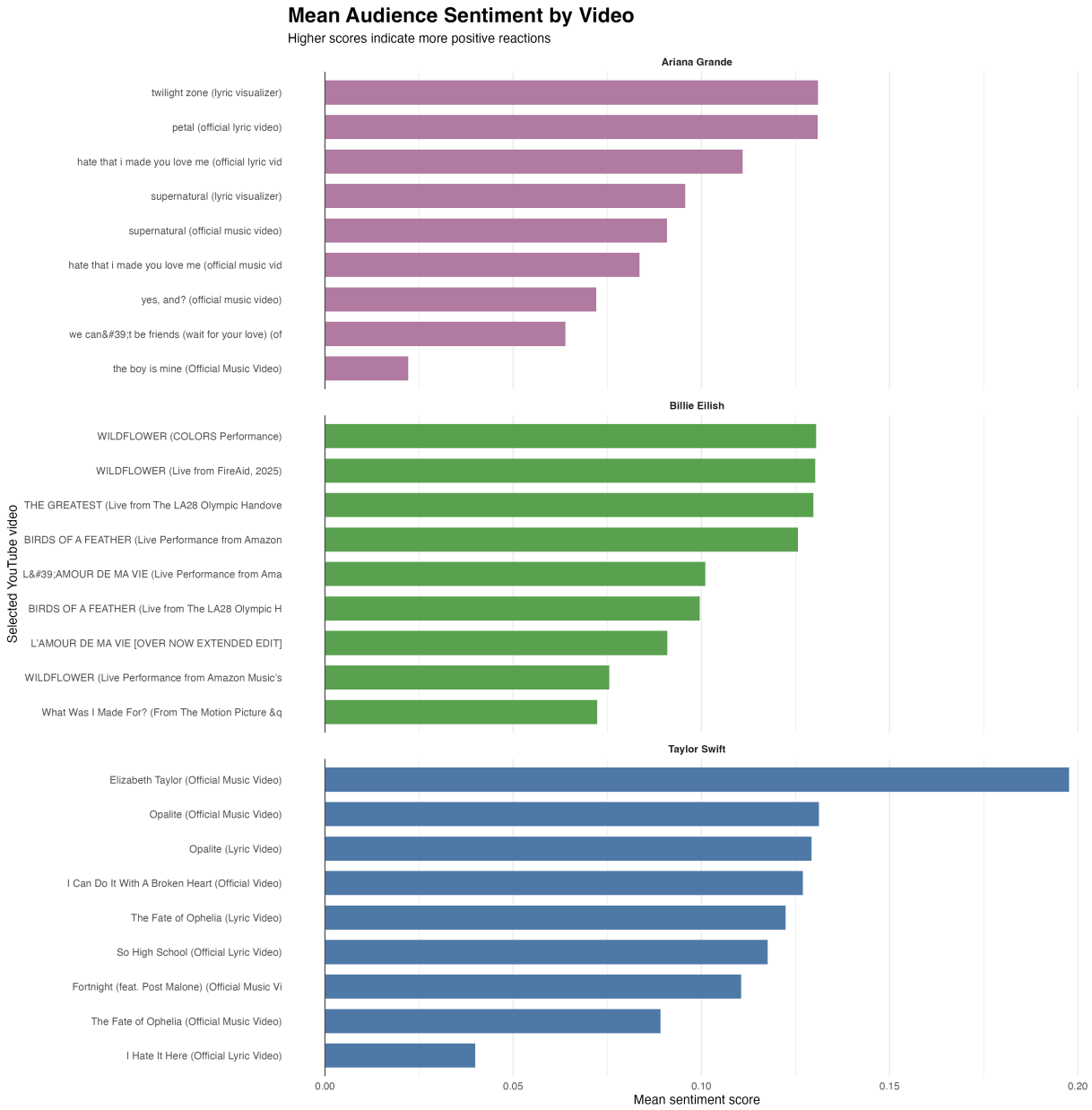


Figure 8b. Saved R mean-sentiment plot for all 27 videos. Labels are abbreviated in the original export; HTML entities represent apostrophes or quotation marks.

Taylor ranges from **0.0399** for *I Hate It Here* to **0.1977** for *Elizabeth Taylor*. Ariana ranges from **0.0221** for *the boy is mine* to **0.1310** for *twilight zone*; *petal* is effectively tied with *twilight zone* at the plotted precision. Billie has a narrower range: **0.0723** for *What Was I Made For?* (Barbie Montage) to **0.1305** for *WILDFLOWER* (COLORS).

All 27 means exceed zero, although every artist sample includes negative comments. Different video formats and lyrical content may contribute to variation, but these descriptive scores do not establish a cause or statistical significance. Interpret the bars with the positive/negative shares in Table 17 and the original text.

10. Decision Tree

The task is to predict which artist a comment concerns from its cleaned words: Taylor Swift, Ariana Grande or Billie Eilish. Artist and video identifiers are excluded as predictors. The corrected Q10 script was executed successfully in R 4.6.0; the results below replace the earlier test scores.

Training data and predictors

With seed 3032, the script selects 80% of comments within each artist: 8,640 training records (2,880 per class) and 2,160 test records (720 per class). The training texts supply the 250 most frequent terms of at least three characters. Predictors are binary: 1 if a term occurs, otherwise 0. Test columns are explicitly reordered to the training term names before assigning model-variable names. Assertions check that both orders match.

Baseline and improvement attempt

Both models use rpart classification trees. The baseline uses cp = 0.01, minsplit = 30, minbucket = 10 and maxdepth = 8, with xval = 10. The tuned model permits smaller leaves and a deeper tree: minsplit = 20, minbucket = 7 and maxdepth = 20. These changes aim to retain useful distinctions while controlling overfitting.

Five-fold cross-validation on the training data selects by accuracy from cp = {0.0005, 0.001, 0.002, 0.005, 0.01, 0.02}. The selected value remains 0.0005. The final tuned model is fitted on the training set, with internal rpart cross-validation disabled (xval = 0). Both models use the same held-out test records. Because the attempt changes several tree controls, its effect cannot be attributed to cp alone.

Model	Accuracy	Balanced accuracy	Macro precision	Macro recall	Macro F1
Baseline tree	0.3546	0.3546	0.4469	0.3546	0.2094
Tuned tree	0.3889	0.3889	0.5593	0.3889	0.2928

Table 19. Corrected test-set performance.

Artist	Precision	Recall	F1	Support
Taylor Swift	0.3573	0.9597	0.5207	720
Ariana Grande	0.7679	0.1194	0.2067	720
Billie Eilish	0.5526	0.0875	0.1511	720

Table 20. Corrected tuned-tree performance by class.

Actual artist	Predicted Taylor	Predicted Ariana	Predicted Billie
Taylor Swift	691	7	22
Ariana Grande	605	86	29
Billie Eilish	638	19	63

Table 21. Corrected tuned-tree confusion matrix.

10. Decision Tree: Results and Interpretation

Tuning raises accuracy from 35.46% to 38.89% (+3.43 percentage points) and macro F1 from 0.2094 to 0.2928. The tuned accuracy is 5.56 points above always predicting one class in this balanced test set. The model still predicts Taylor for 1,934 of 2,160 comments (89.54%). Taylor recall is 95.97%, compared with 11.94% for Ariana and 8.75% for Billie.

Accuracy counts all correct predictions. Macro precision, recall and F1 give each class equal weight. Balanced accuracy is mean class recall and equals accuracy here because each class has 720 test records. The confusion matrix shows that improved accuracy still leaves strong Taylor prediction bias. These results do not support using the classifier for reliable automatic artist identification.

Term	Relative importance	Term	Relative importance
good	100.00	ari	73.72
travis	94.79	supernatural	71.85
elizabeth	87.29	eyes	69.86
performance	80.68	voice	66.16
grammy	74.88	sunshine	61.42

Table 22. Ten most important terms; the largest importance is rescaled to 100.

Terms such as elizabeth, travis, ari and supernatural can identify artist-specific discussion, while good, voice and performance are shared evaluative words. Importance describes a term within the fitted tree; it is neither a causal effect nor a percentage of prediction accuracy. The importance ranking and selected cp remain unchanged after correcting test-column alignment.

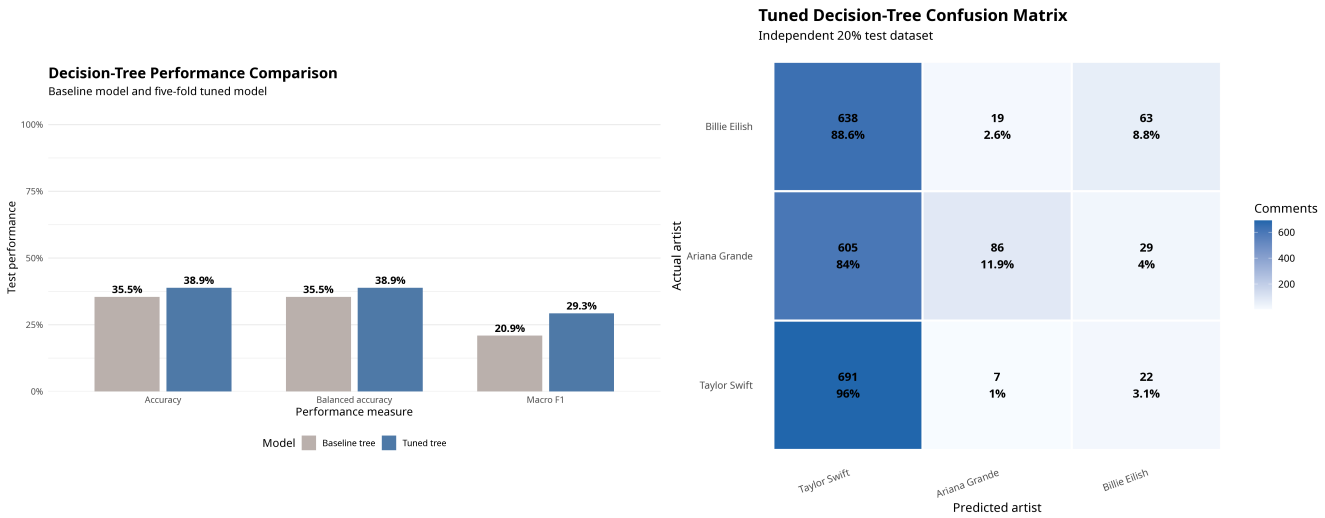


Figure 9. Regenerated R plots: corrected model comparison and tuned-tree confusion matrix.

```
print(q10_model_comparison_plot)
print(q10_confusion_plot)
```

Validation scope and limitations

The earlier term-order defect has been corrected and the complete Q10 script rerun. The figures and saved CSV/RData outputs now use aligned test features. Vocabulary selection still occurs before cross-validation, so fold scores are not fully isolated from vocabulary selection. The comment-level split does not group identical text, authors or videos. Repeated lyrics, shared vocabulary and sparse short comments may also limit generalisation. A future improvement would fit vocabulary within each training fold and evaluate on grouped, unseen videos or authors.

11. Topic Modelling

Latent Dirichlet Allocation (LDA) fitted three topics separately to each artist dataset. Documents empty after the modelling-stage preprocessing were excluded. Using three topics provides a consistent display; topic numbers do not identify matching themes across artists.

Artist	Original docs	Docs used	Empty removed	Terms	Tokens	Log likelihood	Perplexity
Taylor Swift	3600	2609	991	699	9724	-53649.45	426.18
Ariana Grande	3600	2625	975	1034	17020	-97936.12	489.70
Billie Eilish	3600	2448	1152	668	12333	-64852.20	303.42

Table 23. LDA model summary

Artist	Topic	Mean probability	Dominant docs	Top terms
Taylor Swift	1	0.333	1061	life, happy, time, feel, every, great, hate, year, album, anyone
Taylor Swift	2	0.334	867	love, thank, que, always, hope, amazing, new, better, please, day
Taylor Swift	3	0.333	681	best, never, beautiful, ever, elizabeth, world, back, married, forever, heart
Ariana Grande	1	0.332	897	life, god, years, back, always, help, david, time, day, never
Ariana Grande	2	0.337	1198	petal, ari, believe, best, anyone, beautiful, album, cause, que, happened
Ariana Grande	3	0.332	530	love, mind, supernatural, yeah, come, good, wait, hate, need, loves
Billie Eilish	1	0.338	1378	beautiful, que, voice, amazing, anyone, best, performance, movie, thank, version
Billie Eilish	2	0.331	514	love, time, life, back, good, need, feel, said, mind, moved
Billie Eilish	3	0.331	556	love, til, day, better, live, die, always, might, never, forever

Table 24. Topic prevalence and top terms

Comparison	Shared terms	Shared top-topic terms
Taylor vs Ariana	13	life, time, hate, album, anyone, love, que, always, day, best, never, beautiful, back
Taylor vs Billie	16	life, time, feel, anyone, love, thank, que, always, amazing, better, day, best, never, beautiful, back, forever
Ariana vs Billie	14	life, back, always, time, day, never, best, anyone, beautiful, que, love, mind, good, need

Table 25. Topic-term overlap between artists

Each document has a probability for every topic, so mean topic probabilities near one third can coexist with unequal dominant-document counts. A dominant label is just the largest probability; it need not be a confident assignment. Using three topics for all artists makes the display consistent but does not establish that three is the best model size.

The labels are interpretations of word groups. Taylor includes life/feeling language, appreciation and song-related terms. Billie includes performance praise and lyric-related life/love/death language. Ariana's first topic includes god and david from repeated off-topic posts; it should not be treated as a general fan belief or a musical theme.

The overlaps of 16 terms for Taylor-Billie, 14 for Ariana-Billie and 13 for Taylor-Ariana refer to the listed top words, not equivalent topics. Topic 1 in one independently fitted model has no automatic correspondence to Topic 1 in another.

The perplexities (Taylor 426.18, Ariana 489.70 and Billie 303.42) were calculated on different corpora and vocabularies, so their ordering cannot establish which model is better. Model selection would require candidate models evaluated on the same corpus with a consistent held-out procedure; these results do not provide that comparison.

The models retain 2,609 Taylor, 2,625 Ariana and 2,448 Billie documents: 72.47%, 72.92% and 68.00% of their original samples. The topics describe these retained subsets, not every collected comment. Topic terms remain exploratory because short text, lyrics and repeated off-topic passages affect the inferred groups.

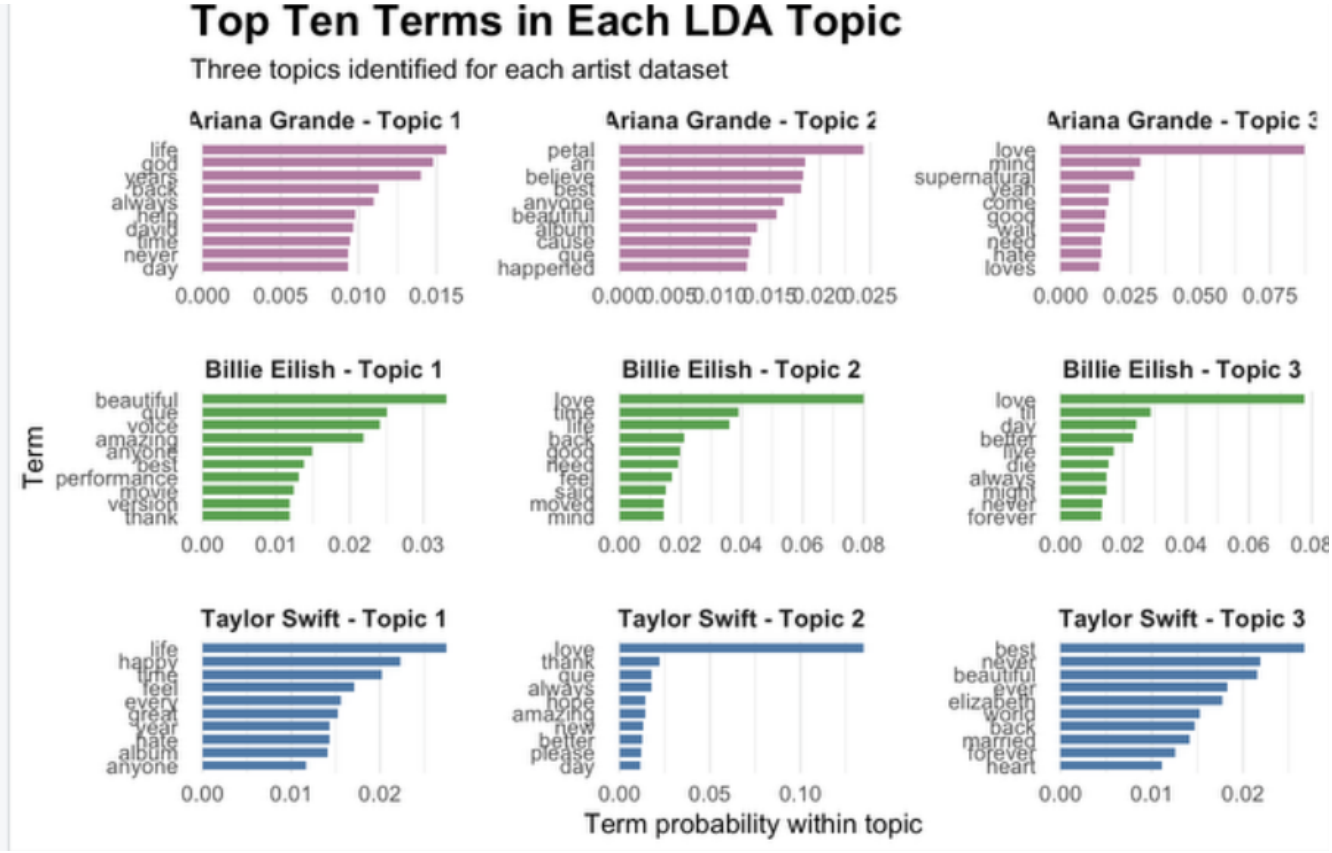


Figure 10. Original LDA topic-term plot, enlarged and cropped to the plotting pane.

```
763
764 # ------------------------------------------------------------
765 # 12. Display final results
766 # ------------------------------------------------------------
767
768 print(q11_model_summary)
769
770 print(
771   q11_topic_summary[, c(
772     "artist",
773     "topic",
774     "mean_topic_probability",
775     "dominant_documents",
776     "top_terms"
777   )],
778   row.names = FALSE
779 )
780
781 print(q11_term_overlap)
782
783 print(q11_topic_terms_plot)
```

References: Methods and Software

Q1 sources are consolidated with the catalogue audit on the following page.

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Q1 Source Notes and References

Catalogue-count audit

The Table 1 totals use the released-song lists consulted on 22 September 2026. The inventory includes distinct released titles under the stated rule; it is not a label-certified lifetime total. Wikipedia supplies the catalogue index, while the label and award organisations support release and achievement claims.

Taylor: 294 entries minus six writing/backing-only entries and five repeated versions = **283**. Exclusions: "1 Step Forward, 3 Steps Back," "Bein' With My Baby," "Best Days of Your Life," "Deja Vu," "This Is What You Came For" and "You'll Always Find Your Way Back Home." "Better Man" remains because Swift released her own version. Alternate versions of "All Too Well," "Bad Blood," "Karma," "Long Live" and "Lover" are merged.

Ariana: 210 entries minus eight alternate versions and the backing-vocal appearance on "Haunt You," plus eleven missing *Petal* tracks = **212**. "Hate That I Made You Love Me" was already listed. Merged versions concern "7 Rings," "34+35," "All My Love," "Into You," "Intro (End of the World)," "One Last Time," "Supernatural" and "Yes, And?" The two different tracks called "Intro" remain separate.

Billie: 66 distinct entries, including named spoken tracks and artist-published early songs. A guest remix remains when it is the artist's only released performance of that title; a live-only cover counts once. Catalogue indexes can omit releases, so totals describe this documented inventory rather than every possible counting convention.

References supporting Q1

Ariana Grande. (n.d.). In *Wikipedia*. Retrieved September 22, 2026, from https://en.wikipedia.org/wiki/Ariana_Grande

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12. Dashboard - Page 1: Audience Overview

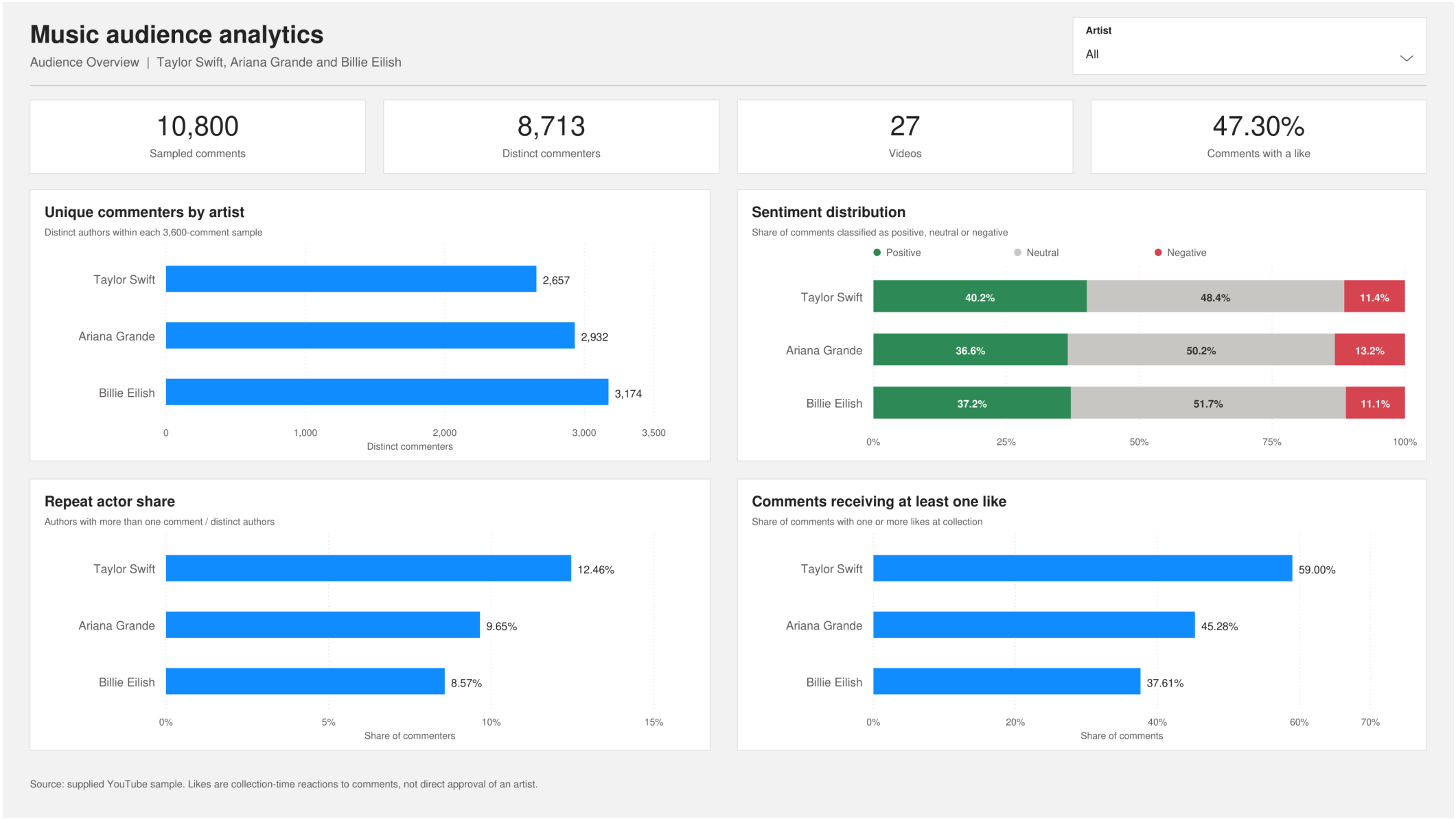


Figure 11. Audience breadth, sentiment, repeat participation and comments receiving likes.

Key finding: Billie has the widest sampled participation, while Taylor has the highest repeat-author share and the largest share of comments receiving likes.

Analytical preview rendered from the supplied data; not a Power BI Desktop screenshot. Chart explanations and functionality are on pages 29-31.

12. Dashboard - Page 2: Participation & Themes

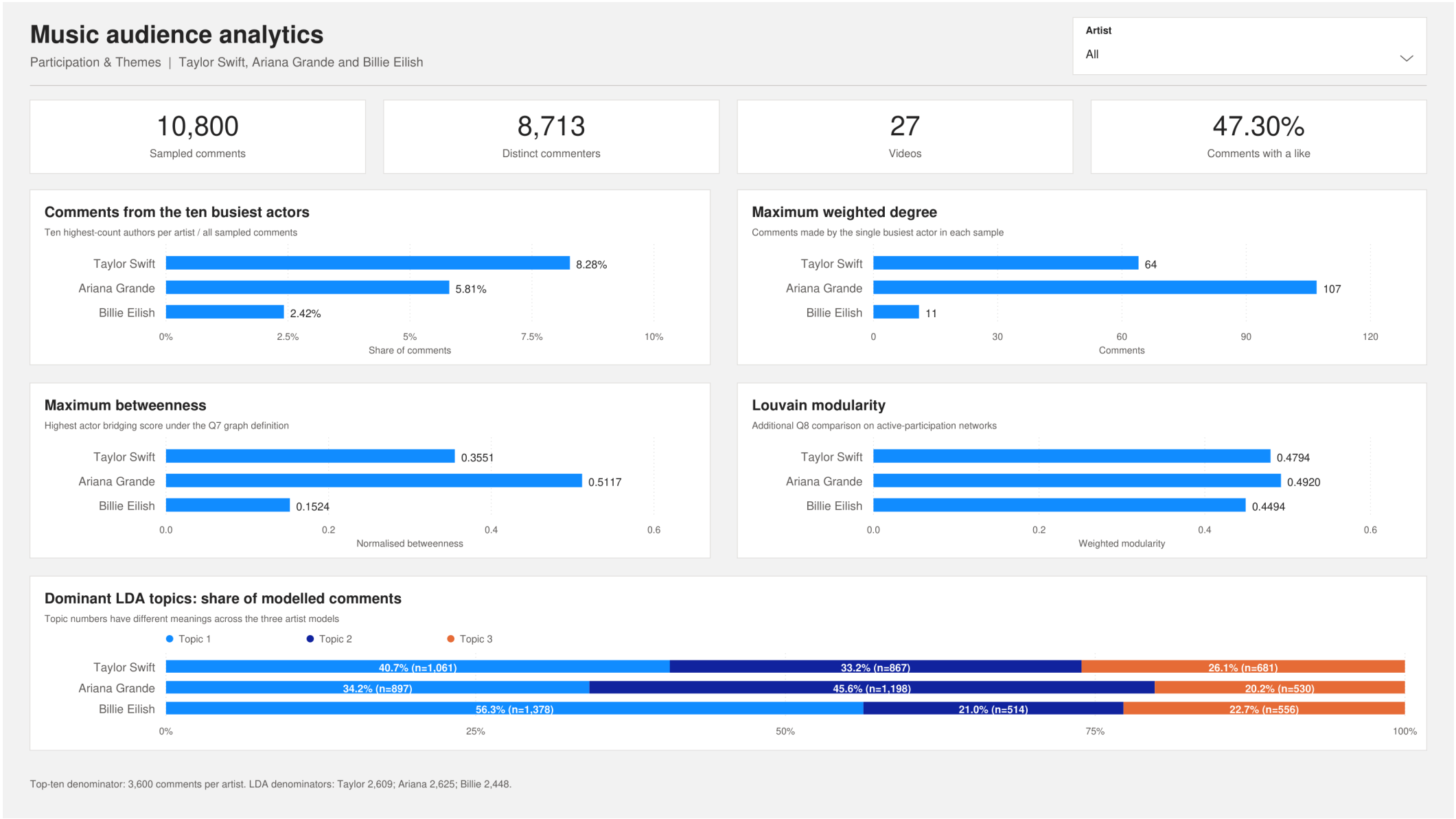


Figure 12. Comment concentration, actor centrality, community separation and dominant topics.

Key finding: Ariana has the busiest individual author (107 comments), but Taylor's ten busiest authors contribute a larger share collectively (8.28% versus 5.81%).

Analytical preview rendered from the supplied data; not a Power BI Desktop screenshot. Chart explanations and functionality are on pages 29-31.

12. Dashboard: Audience Overview

The Power BI project contains two report pages with nine analytical charts: four on Audience Overview and five on Participation & Themes. Each page also has four summary cards and an artist selector. The charts use the existing YouTube sample and R exports. Figures 11 and 12 show analytical previews of the layout; they are not captures from Power BI Desktop.

Purpose and design

The dashboard brings audience breadth, repeated participation, sentiment and network structure into one view for a label manager. Horizontal bars make the three artist names and values easy to compare. Percentage charts use a common denominator within each measure. Neutral backgrounds, direct data labels and consistent colours keep the focus on the results. Positive, neutral and negative sentiment have distinct colours.

1. Unique commenters by artist

This chart counts distinct author channel IDs in each 3,600-comment sample. The bars show that Billie has the broadest sampled participation (3,174 authors), followed by Ariana (2,932) and Taylor (2,657). It was included because comment totals alone hide repeated posting. A manager can compare breadth with the repeat-author chart before interpreting activity as a large or deeply engaged audience. These figures describe the selected comments, not each artist's total fan base.

2. Sentiment distribution

A 100% stacked bar shows the positive, neutral and negative shares for each artist. Taylor has the highest positive share (40.17%), compared with Billie (37.17%) and Ariana (36.58%). The composition view was chosen to show all three categories together. A manager can identify samples for closer reading, including Ariana's 13.19% negative share. Negative words can occur in quoted lyrics or personal stories, so the chart is a screening tool rather than evidence of criticism.

3. Repeat actor share

This is the number of authors who posted more than once divided by distinct authors: Taylor 12.46%, Ariana 9.65% and Billie 8.57%. It complements the breadth chart by showing repeated participation. A manager could review the recurring conversations in Taylor's sample while considering ways to encourage return participation in Billie's. Repeated posting does not, by itself, demonstrate loyalty, influence or positive sentiment.

4. Comments receiving at least one like - new analysis

This chart divides comments with like_count greater than zero by all sampled comments. Taylor has 59.00%, Ariana 45.28% and Billie 37.61%. It was added to distinguish visible reactions from the tone of comment text. A manager can inspect which comments attract reactions before designing further engagement activity. The measure concerns likes on comments; it is not video engagement rate or direct approval of the artist. Comment age and exposure differ.

12. Dashboard: Participation & Themes

5. Comments from the ten busiest actors - new analysis

The chart shows comments written by the ten highest-count authors as a share of each artist's 3,600 comments. Taylor's top ten contribute 298 comments (8.28%), Ariana's 209 (5.81%) and Billie's 87 (2.42%). It was included to assess concentration beyond the single busiest author. A manager should check this alongside audience breadth before treating frequent voices as representative of the whole sample. These authors are selected by comment count, not PageRank.

6. Maximum weighted degree

In the author-video networks, weighted degree is the number of comments made by an author. The largest value is 107 for Ariana, 64 for Taylor and 11 for Billie. This chart was chosen to reveal exceptional individual activity. A manager can compare it with the top-ten chart to see whether a pattern depends on one account or several active participants. High frequency alone does not establish spam or social influence.

7. Maximum betweenness

This chart compares the largest actor betweenness score from Q7: Ariana 0.5117, Taylor 0.3551 and Billie 0.1524. It was included to show structural bridging that comment volume alone cannot describe. A manager could inspect the relevant author's participation across videos to understand which sampled discussions are connected. The values depend on the graph definition and path weights; they do not demonstrate that the author actually spreads information between people.

8. Louvain modularity

The common scale compares modularity from the additional Louvain analysis: Ariana 0.4920, Taylor 0.4794 and Billie 0.4494. It helps a manager consider whether discussion groups merit separate inspection. These are the restricted active-participation networks from Q8, not networks of every sampled author. A larger value suggests stronger separation under the chosen model, but does not prove more meaningful fan groups. Girvan-Newman remains the required Q8 method; Louvain supplies a comparison.

9. Dominant LDA topics: share of modelled comments

The 100% stacked bars show the share of retained documents assigned to each dominant topic. Their denominators are 2,609 for Taylor, 2,625 for Ariana and 2,448 for Billie. This view was chosen to compare within-artist topic concentration despite different retained totals. Billie has 56.3% of dominant assignments in Topic 1, but its mean topic probabilities remain near one-third, so dominance should not be read as strong topic separation. The project includes artist-specific topic words in tooltips. A manager should review those words and representative comments before using a theme in content planning. Topic numbers are not equivalent across artists.

Reading the cards

With all artists selected, the cards show 10,800 comments, 8,713 distinct commenters, 27 videos and 47.30% of comments receiving a like. The distinct-commenter total is calculated across all records; summing the three artist totals would double-count authors appearing in more than one artist sample.

12. Dashboard: New Insights, Functionality and Checks

Two findings beyond the earlier analyses

Concentration depends on the level examined. Ariana has the busiest individual author (107 comments versus Taylor's 64), yet Taylor has the larger top-ten total (298 versus 209). Ariana's busiest author accounts for 51.2% of its top ten's comments, compared with 21.5% for Taylor. A manager should distinguish one exceptional contributor from activity spread across several frequent participants, then read a broader range of voices before acting on repeated requests.

Positive language and visible reactions have different rankings. Ariana has more comments receiving likes than Billie (45.28% versus 37.61%), although Billie has a slightly larger positive-comment share (37.17% versus 36.58%). The small sentiment difference is not evidence of statistical significance. Managers can use the two measures together to decide which comments to inspect: positive wording and reaction activity answer different questions.

Configured Power BI functionality

Both pages contain an Artist dropdown linked to the same Artists table and synchronisation group. The intended workflow is to select one artist, inspect its cards and charts, move to the other page and clear the selection to compare all three. Native page tabs provide navigation. Data labels show values directly, while tooltips provide supporting values and topic terms. Artist relationships connect the dimension to each artist-level dataset.

Filtering is defined at artist level. The sentiment and LDA tables are aggregate R exports, so selecting a sentiment category or topic does not identify or filter individual comment records by that classification. Adding such a feature would require comment-level classifications and a different model.

Expected card values for the artist filter

Selection	Comments	Distinct authors	Videos	With a like
All artists	10,800	8,713	27	47.30%
Taylor Swift	3,600	2,657	9	59.00%
Ariana Grande	3,600	2,932	9	45.28%
Billie Eilish	3,600	3,174	9	37.61%

Calculation and verification scope

The top ten are ranked separately for each artist by descending comment count, with channel ID breaking ties. Like share counts records with like_count > 0. Both use all 3,600 sampled comments per artist. Repeat share uses distinct authors instead. The original 400-comment-per-video cap remains; this dashboard does not recollect or restore replies.

Programmatic checks passed for project structure, field references, source snapshots and the audience calculations. The supporting package includes the CSVs, DAX measures, a repeatable Python verifier and a Power BI validation query. Desktop opening, refresh, native rendering and interactive filtering have not been tested. The previews and file checks therefore do not establish a completed Power BI functionality test.